J.P.Morgan

Global Markets Strategy
13 May 2026

Flows & Liquidity

Revisiting equity supply and demand from the coming IPOs

- **The typical jump in the free float share after the expiry of the lock-up period in IPOs reflects a rather mechanical reclassification by index providers and is not necessarily the result of selling by insiders or other shareholders.**
- **What is certain is that the inclusion of the anticipated big three IPOs into equity indices would necessitate sizeable passive demand from index inclusion and rebalancing.**
- **We estimate $410bn of total passive buying once lock up periods expire, most likely during H1 2026. This $410bn of buying by passive funds could be accommodated by a combination of companies themselves releasing previous repurchased shares or non-strategic investors selling to passive funds.**
- **At index level, the increased weights of the newly included companies would come at the expense of the weights of the companies they replace as well as the biggest tech companies like Mag7. That said, this index rebalancing would not necessarily entail actual selling of Mag7 stocks by passive funds if the past years pattern of strong and steady inflows into passive equity funds continues going forward.**
- **The YTD rotation away from Chinese tech towards Korea and Taiwan tech continues in the equity futures space, with only tentative signs of a pause in the equity ETF space.**
- **The underperformance trend of both ethereum and other altcoins vs. bitcoin is unlikely to change unless we see meaningful improvements in network activity, DeFi and real world applications.**

Global Markets Strategy

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See page 28 for analyst certification and important disclosures.

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Cross Asset Fund Flow Monitor

Current level shows the latest percentile of weekly flows; Min is denoted by 0 and Max by 1. As of 6th May 26.

MF & ETF Flows	Min	Max	4 wk avg ($bn)	2025 avg ($bn)
All Equities			11.6	8.1
All Bonds			13.0	11.3
US Equities			8.3	3.5
US Bonds			4.7	4.3
Non-US Equities			3.3	4.6
Non-US Bonds			8.4	7.0
US HG Bonds			3.3	3.4
US HY Bonds			1.2	0.4
US Lev. Loans *			0.3	0.0
US MMFs			-0.3	4.0
EM Equities			0.5	0.9
EM Bonds			1.54	0.50
Japan Equities			0.4	-0.1
China Equities			-0.19	-0.15
Europe				
Europe Equities			-2.0	0.9
Europe Bonds			4.0	4.3
Europe HG Bonds			-0.3	0.9
Europe HY Bonds			0.26	0.17
Europe MMFs			2.5	3.8
Other Equities			4.59	3.00

Source: Lipper, ICI, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.
* US LL historical flows are monthly averages converted to weekly for comparison. China onshore A-share ETFs have been excluded.

Cross Asset Positioning Monitor

Current level shows the latest percentile, Min is denoted by 0 and Max by 1.

As of 12-May-26	MIN	MAX	Current percentile
Equities			0.72
Govt Bonds			0.52
Credit			0.28
Dollar			0.47
Commodities ex Gold			0.90
Gold			0.67
Bitcoin			0.66
EM Equities			0.59
EM Bonds/FX			0.32
Japan Equities			0.90
Europe Equities			0.75
US Equity Sectors:			
Energy			0.78
Materials			0.66
Industrials			0.45
Discretionary			0.59
Staples			0.17
Health Care			0.37
Financials			0.29
Technology			0.79
Communication Services			0.54
Utilities			0.40

Source: J.P. Morgan Flows & Liquidity.
Cross Asset Positioning Monitor aggregates across the various position indicators of Appendix ranging from positioning proxies across various futures contracts, momentum signals as proxies of how trend-following funds/CTAs are positioned, mutual fund betas as proxies of how mutual fund managers are positioned, risk parity fund positioning and leverage proxies, hedge fund betas as proxies of how hedge fund managers are positioned, client surveys, asset allocation estimates of private non-bank investors at global level, short interest indicators, etc.

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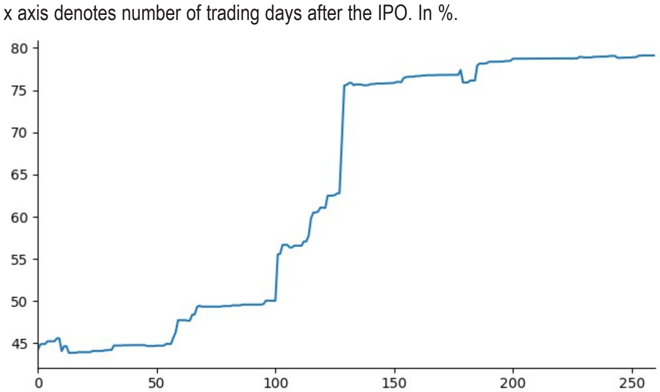
- We have noted previously (F&L, Feb 26th) that the close to zero net equity issuance that underpins our bullish equity demand/supply projection for this year is unlikely to change as a result of the potential IPOs by the three biggest private AI companies. This in turn has raised questions in our conversations over what happens after the companies go public. To look at this question, we look at the experience on the aftermath of companies listing via IPOs in terms of the evolution of the share of free float to total market cap and the holdings of insiders, typically company officers and directors as well as large shareholders that hold a more than 10% stake.
- We start by looking at ten of the largest IPOs since 2010 for data availability on insider shares, and look at the free float share as well as the percentage share of insider holdings in the first 12 months after the IPO. Figure 1 and Figure 2 show the average of the free float shares and the individual free float shares of these IPOs respectively, and Figure 3and Figure 4 show the average of insider shares and the insider share for the individual companies respectively.
- We make the following observations. First, the free float shares normally increase relatively slowly until lockups expire, typically around 6 months after the IPO. Second, there is considerable variation in level in terms of both free float shares and insider shares at the company level, reflecting a significant heterogeneity in capital structures. This suggests that while there is a common trend in the free float share, we cannot generalise from this historical experience on where the coming three large AI company IPOs will settle around in terms of the level of free float and insider shares. And third, there is on average a trend for a modest decline in the insider shares in the year after the IPO, though at the company level there are also significant differences (for many companies there has been little reduction in insider stakes).

Figure 1: Average free float shares

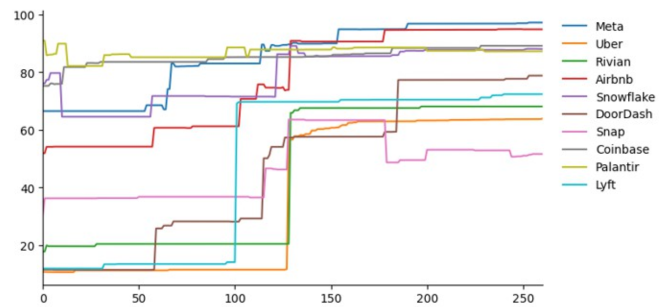
x axis denotes number of trading days after the IPO. In %.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 2: Breakdown of free float shares

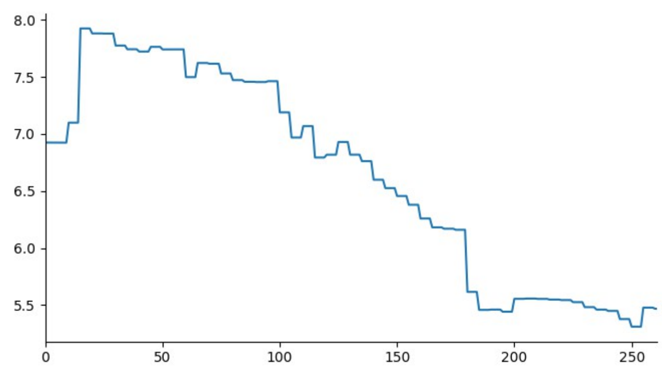
x axis denotes number of trading days after the IPO. In %.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 3: Average of Insiders shares

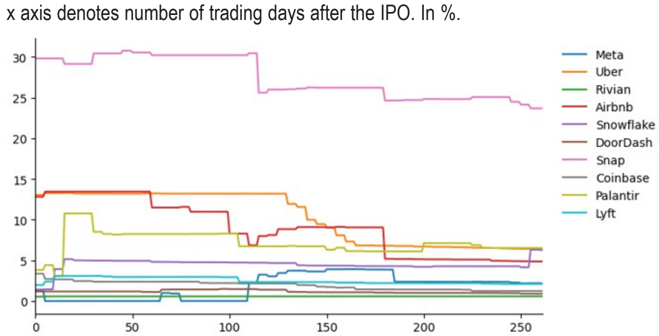
x axis denotes number of trading days after the IPO. In %.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 4: Breakdown of Insiders shares

x axis denotes number of trading days after the IPO. In %.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Is the typical jump in the free float share after the expiry of the lock-up period the result of selling by insiders or other shareholders? The answer to our mind is no. Instead, the jump in the free float share after the expiry of the lock-up period is rather mechanical. This is due to the fact that once the lock-up periods expire, the holdings of non-insiders, i.e. non-strategic investors with a stake of

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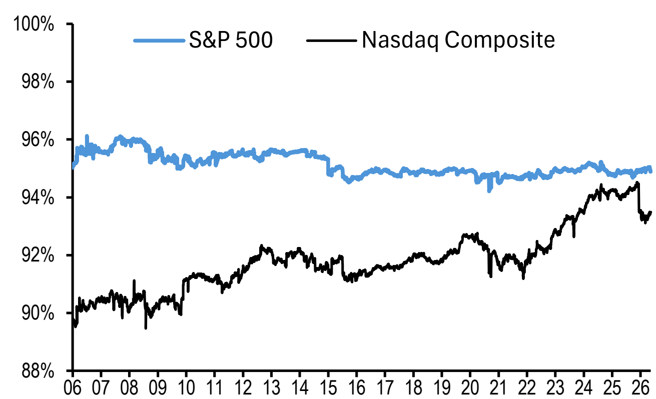
less than 10% or employees other than those classified as insiders, are counted toward the free float whether they sell or keep their stakes. This is effectively a reclassification effect by index providers. An argument could also be made that there have been already several opportunities for non-insiders to sell part of their stakes in the big three private AI companies before listing through structured tender offers or secondary market transactions, though in practice the capacity of these secondary markets may be limited. And companies themselves may have been willing to repurchase some of their shares ahead of the listing, perhaps to release these holdings after the lock-up period expiry to facilitate passive demand.

- Indeed, what is certain is that the inclusion of the anticipated big three IPOs into equity indices would necessitate sizeable passive demand from index inclusion and rebalancing. While the visibility of capital structures is limited, it seems that the free float of these companies would eventually likely increase to around 30%-50% after the 6-month lock up periods expires, most likely during the first half of 2027. How much passive buying could this generate?
- Our colleague Min Moon in equity index research has outlined some timelines of how the inclusion process could take place for different index providers, as well as some calculations of how much passive demand could be created at different levels of free float share assumptions (see [Large Cap IPOs and Potential Index Inclusion], May 11th). Assuming the three anticipated IPOs account for a combined market cap of around $3.5tr, and that at listing around 4% of the shares would be counted toward the free float, this would imply passive rebalancing demand of around $40bn in total when they are initially included in indices. At the end of the lockup periods (typically 6 months) and assuming that the free float share increases to around 40%, or the mid-point of the 30%-50% range, this could create additional passive rebalancing demand of $370bn bringing the cumulative total rebalancing related demand to $410bn. This $410bn of buying by passive funds could be accommodated by a combination of companies themselves releasing previous repurchased shares or non-strategic investors selling to passive funds. At index level, the increased weights of the newly included companies would come at the expense of the weights of the companies they replace as well as the biggest tech companies like Mag7. That said, this index rebalancing would not necessarily entail actual selling of Mag7 stocks by passive funds if the past years pattern of strong and steady inflows into passive equity funds continues going forward as a disproportionate amount of these inflows could be deployed into the newly listed companies to reach the increased weights.
- What about the historical experience at the index level? Figure 5 shows the free float to total market cap ratio for the S&P 500 and Nasdaq composite. The free float share has historically been more stable, which reflects the fact that by the time companies are listed in the S&P they tend to be more mature and established, there are stricter inclusion criteria, and the share of institutional holdings tends to be higher. That said, there has been a greater convergence over time as the free float share of the Nasdaq has been rising towards that of the S&P over time. In part, this is likely due to the maturing of the tech sector and institutionalisation of ownership, as well as increasing overlap as the S&P has become increasingly tech heavy.
- This process has not been linear, however. For example, from 2020 until late 2021, the free float share of the Nasdaq declined significantly amid a tech IPO and SPAC boom that saw a lot of newly public companies with significant stakes from founders as well as PE and VC investors, and perhaps with increased stock rewards for insiders and employees amid the bull market. From 2022 to 2024, this reversed as lockup periods expired, allowing founders to sell to institutional investors and as strategic investor holdings were either sold or reclassified as part of the free float, as the market correction reduced tech weights somewhat and as IPO activity dried up. The inclusion of the anticipated three IPOs into equity indices at a somewhat more expedited timeline could see another decline in index free float ratios as the new companies enter equity indices at lower free float ratios than the current index average.

Figure 5: Free float to total market cap ratios for the S&P 500 and Nasdaq composite

In %.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

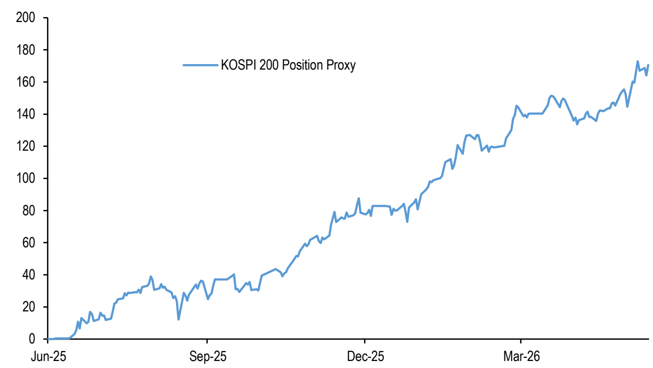
- In all, the typical jump in the free float share after the expiry of the lock-up period in IPOs reflects a rather mechanical reclassification by index providers and is not necessarily the result of selling by insiders or other shareholders. That said the inclusion of the anticipated big three IPOs into equity indices would necessitate sizeable passive demand from index inclusion and rebalancing. We estimate $410bn of total passive buying once the lock-up periods expire, most likely during H1 2026. This $410bn of buying by passive funds could be accommodated by a combination of companies themselves releasing previously repurchased shares or non-strategic investors selling to passive funds. At index level, the increased weights of the newly included companies would come at the expense of the weights of the companies they replace, as well as the biggest tech companies like Mag7. That said, this index rebalancing would not necessarily entail actual selling of Mag7 stocks by passive funds if the past years pattern of strong and steady inflows into passive equity funds continues going forward.

The YTD rotation away from Chinese tech towards Korea and Taiwan tech continues in the equity futures space with only tentative signs of a pause in the equity ETF space

- One of the most intense flows YTD has been the inflow impulse into Korean and Taiwanese equities. This strong flow impulse has been seen in both the futures and equity ETF spaces. This is shown by Figure 6 to Figure 9. Figure 6 and Figure 7 depict our futures positioning proxies based on open interest changes for KOSPI and TAIEX/FTSE Taiwan equity futures. Figure 8 and Figure 9 depict our cumulative flows as % of AUM for Korean and Taiwanese offshore equity ETFs. Overall, Figure 6 and Figure 9 suggest that the acceleration of inflows into Korean and Taiwanese equities over the past five months has not only been driven by futures based investors such as hedge funds but also offshore retail investors via offshore ETFs.

Figure 6: Our futures positioning proxy for Korean equity futures

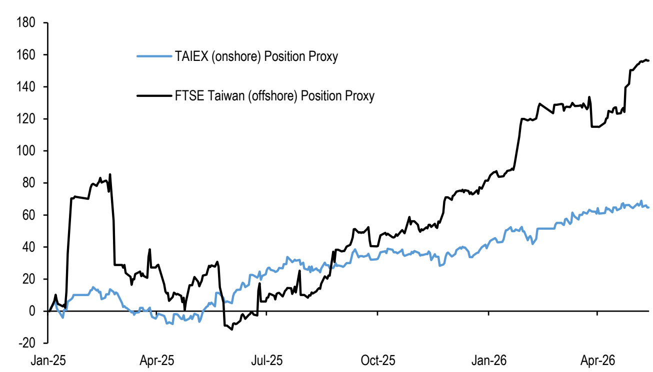
In 000's of contracts.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 7: Our futures positioning proxy for Taiwanese equity futures

In 000's of contracts.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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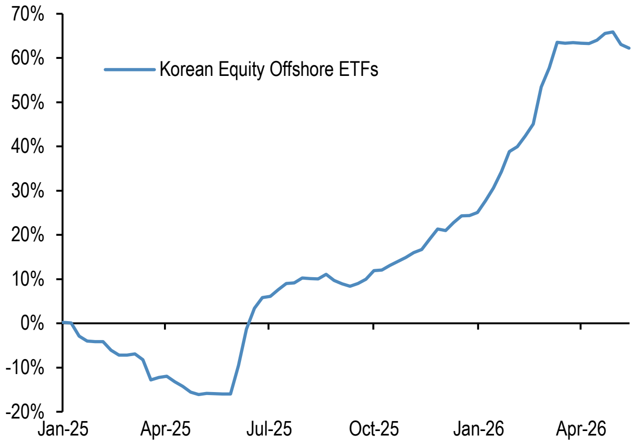
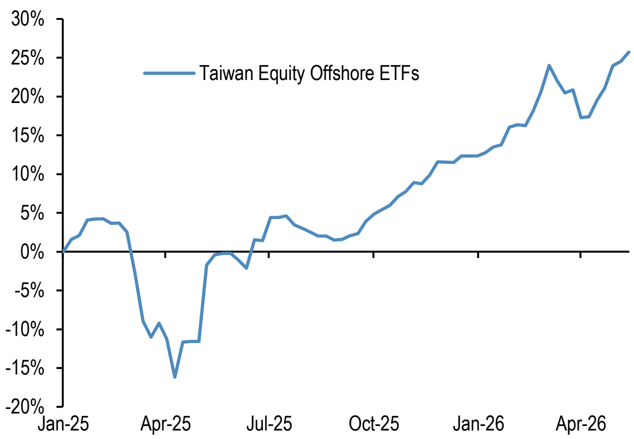
Figure 8: Cumulative flows as % of AUM into offshore Korea equity ETFs



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

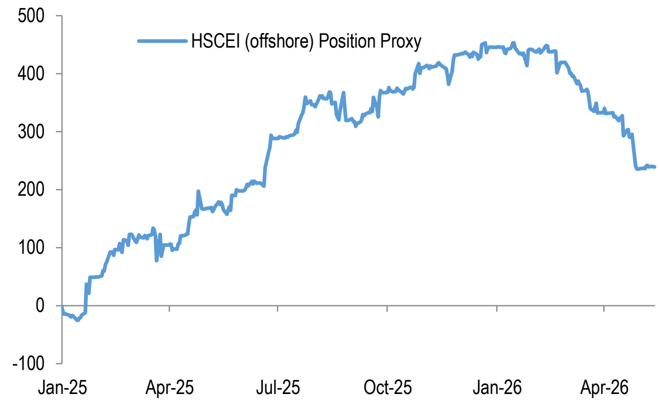
Figure 9: Cumulative flows as % of AUM into offshore Taiwan equity ETFs



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

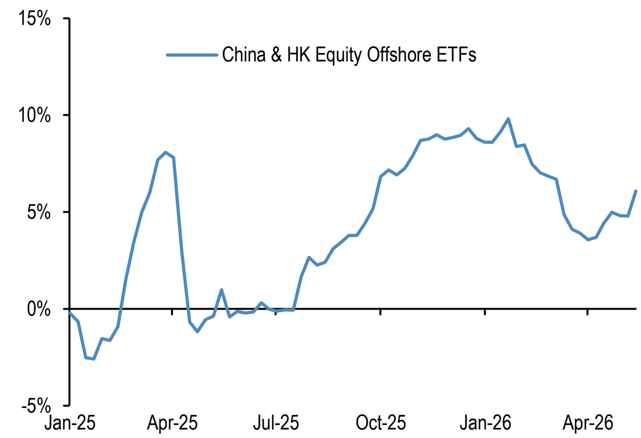
Figure 10: Our futures positioning proxy for HSCEI futures

In 000's of contracts.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 11: Cumulative flows as % of AUM into offshore China & HK equity ETFs



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- What has been equally striking this year is that the acceleration of inflows into Korean and Taiwanese equities has come at the expense of China tech, with our futures positioning proxy for HSCEI futures on a downward trajectory YTD. This is a sharp reversal from the steep upward trajectory in HSCEI futures positions of the previous two years (Figure 10). A broader investor shift towards upstream AI companies might have been one reason behind this rotation away from Chinese tech towards Korea and Taiwan tech and/or simply an increase in concerns about the monetization strategies of Chinese AI companies. So far there is little sign that this rotation is abating in the futures space (Figure 6,Figure 7,Figure 10) with only tentative signs of a pause in the equity ETF space (Figure 8 and Figure 11).

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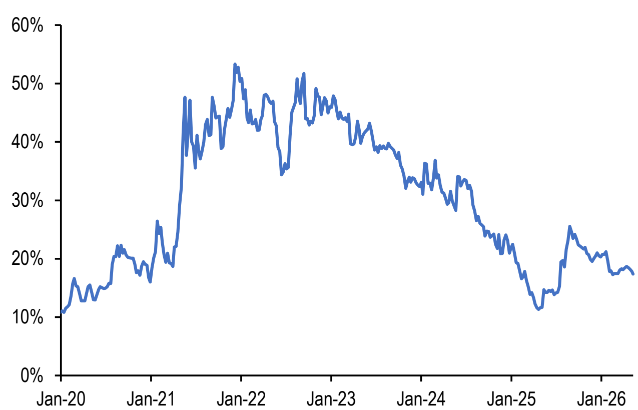
The underperformance trend of both ethereum and other altcoins vs. bitcoin is unlikely to change unless we see meaningful improvements in network activity, DeFi and real world applications

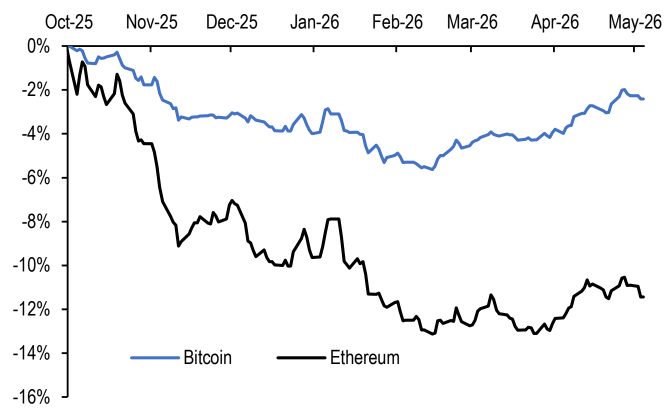
- Despite the crypto market recovery since the Iran conflict erupted, ethereum underperformed bitcoin as shown in Figure 12. This underperformance is also seen in the ETF flow space. While spot bitcoin ETFs have recovered roughly two-thirds of their previous outflows, spot ethereum ETFs have barely recovered a third (Figure 13). Looking at CME bitcoin and ethereum futures as a proxy for institutional positioning, we see a similar picture, with bitcoin futures positioning having recovered almost all of its previous drawdown (Figure 14), while ethereum futures positioning is still some way from fully recovering (Figure 15).

Figure 12: Ratio of ethereum to bitcoin's marketcap



Source: CoinMarketcap, J.P. Morgan Flows & Liquidity.

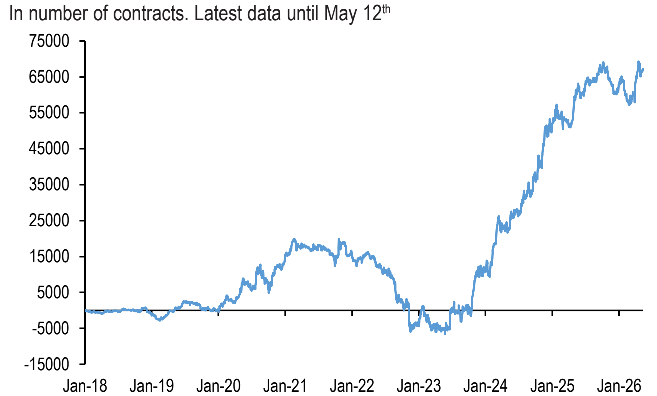
Figure 13: Spot bitcoin and ethereum flow as % AUM from Oct10th 2025 onwards



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 14: Our position proxy based on open interest changes in CME bitcoin futures contracts

In number of contracts. Latest data until May 12th



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 15: Our position proxy based on open interest changes in CME ethereum futures contracts

In number of contracts. Latest data until May 12th

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- There is somewhat less differentiation with momentum

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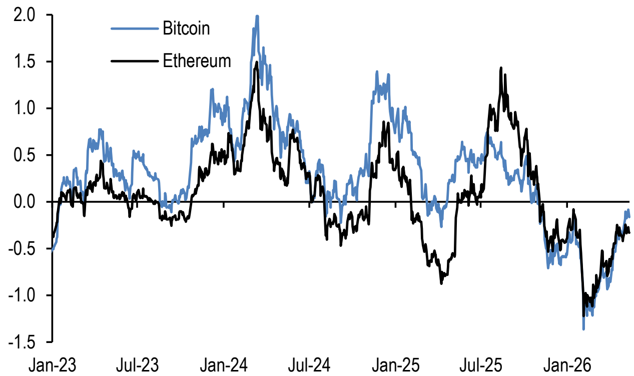
traders however as on our calculations they still appear to be slightly underweight both bitcoin and ethereum (Figure 16), suggesting that CTAs and crypto quant funds have yet to rebuild significant long positions since the October 10th 2025 deleveraging event.

Figure 16: Our proxy for momentum traders positioning on bitcoin and ethereum



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Could upcoming ethereum upgrades boost its price relative to bitcoin? Historically, upgrades have not translated into stronger on-chain activity; instead, they have reduced Layer 2 costs, leading to lower network fees and diminished token burn, which in turn increased net supply and weakened price support. The upcoming 2026 upgrades (Glamsterdam and Hegota) aim to raise network throughput and lower transaction costs on ethereum chain, strengthening ethereum's scalability and broader utility. However, it remains to be seen whether these coming improvements can succeed in boosting network activity or at least drive enough demand growth to offset the continued reduction in ethereum's burn mechanism and resulting net supply increase. As mentioned in our previous publications, ethereum upgrades over the previous three years had failed to succeed in boosting network activity.
- Altcoins ex ethereum have been similarly underperforming bitcoin since 2023 (Figure 17) because of more adverse liquidity conditions, lower market depth and breadth, lack of DeFi growth, hacks and security breaches. All these factors have eroded confidence in the broader altcoin ecosystem and discouraged the deployment of fresh capital.

Figure 17: Ratio of altcoins (ex ethereum) to bitcoin's marketcap

Source: CoinMarketcap, J.P. Morgan Flows & Liquidity.

- In all, both ethereum and other altcoins have been underperforming bitcoin despite the crypto market recovery since the Iran conflict. And this underperformance trend that started in 2023 is unlikely to change unless we see meaningful improvements in network activity, DeFi and real world applications.

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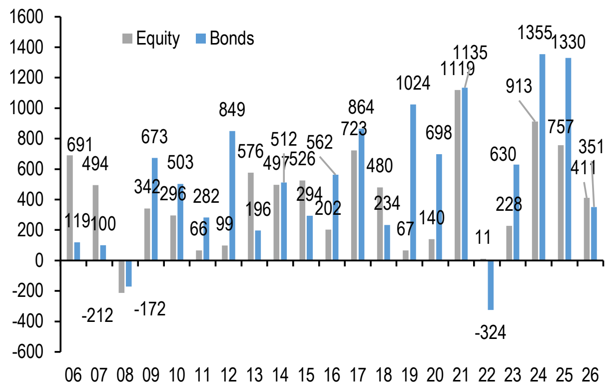
Appendix

Chart A1a: Global equity & bond fund flows

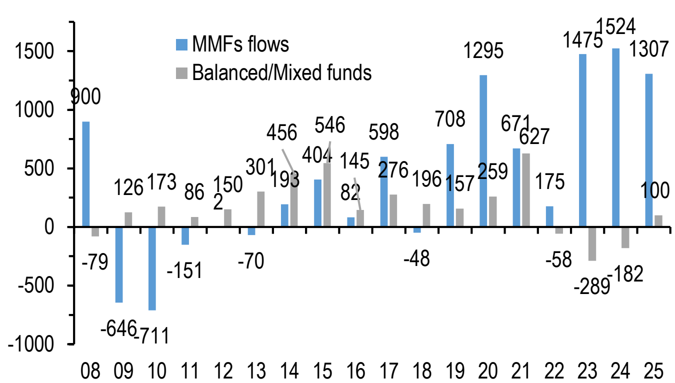
$bn per year of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Flows come from ICI (worldwide data up to Q4'25). Data since then are a combination of monthly and weekly data from Lipper and Bloomberg.



Source: ICI, LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A1b: Quarterly Balanced/Mixed funds & MMFs flows worldwide

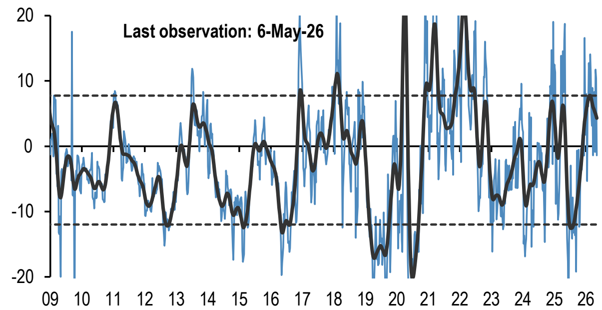
$bn per quarter of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Data come from ICI (worldwide data) and are till Q4'25.



Source: ICI, J.P. Morgan Flows & Liquidity.

Chart A2: Fund flow indicator

Difference between flows into Equity and Bond funds: $bn per week. Difference between flows into Equity vs. Bond funds in $bn per week. Flows include Mutual Fund and ETF flows globally, i.e. funds domiciled both inside and outside the US (source: Lipper). The thin blue line shows the 4-week average of difference between Equity and Bond fund flows. Dotted lines depict ±1 StDev of the blue line since 2009. The thick black line shows a smoothed version of the same series. The smoothing is done using a Hodrick-Prescott filter with a Lambda parameter of 100. China onshore A-share ETFs have been excluded.



Source: LSEG Lipper, J.P. Morgan Flows & Liquidity.

Table A2: Trading turnover monitor

Volumes are monthly and Turnover ratio is annualised (monthly trading volume annualised divided by the amount outstanding). UST Cash is primary dealer transactions in all US government securities. UST futures are from Bloomberg Finance L.P. JGBs are OTC volumes in all Japanese government securities. Bunds, Gold, Oil and Copper are futures. Gold includes Gold ETFs. Min-Max chart is based on Turnover ratio. For Bunds and Commodities, futures trading volumes are used while the outstanding amount is proxied by open interest. The diamond reflects the latest turnover observation. The thin blue line marks the distance between the min and max for the complete time series since Jan-2005 onwards. Y/Y change is change in YTD notional volumes over the same period last year.

As of Mar-26	MIN	MAX	Turnover ratio	Vol (tr)	y/y chng
Equities					
EM Equity			1.1	$1.6	104%
DM Equity			1.5	$14.5	52%
Govt Bonds					
UST cash			12.3	$20.3	7%
UST futures			0.9	$23.6	18%
JGBs			44.5	¥4,727	5%
Bund futures			1.6	€14.9	18%
Credit					
US HG			1.2	$1.0	17%
US HY			0.6	$0.2	3%
US Convertibles			3.8	$0.04	18%
Commodities					
Gold			88.7	$2.9	109%
Oil			99.7	$6.1	-3%
Copper			3.0	$0.9	106%
Digital Assets					
CME Bitcoin			112.3	$0.074	-33%
CME Ethereum			103.1	$0.033	60%

* Data with one month lag

Source: Bloomberg Finance L.P., Federal Reserve, Trace, Japan Securities Dealer Association, WFE, J.P. Morgan Flows & Liquidity.

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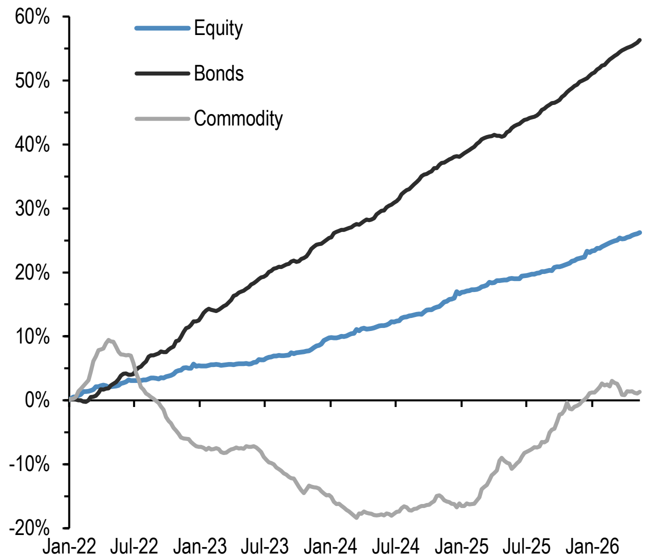
ETF Flow Monitor (as of 13th May)

Chart A3: Global Cross Asset ETF Flows

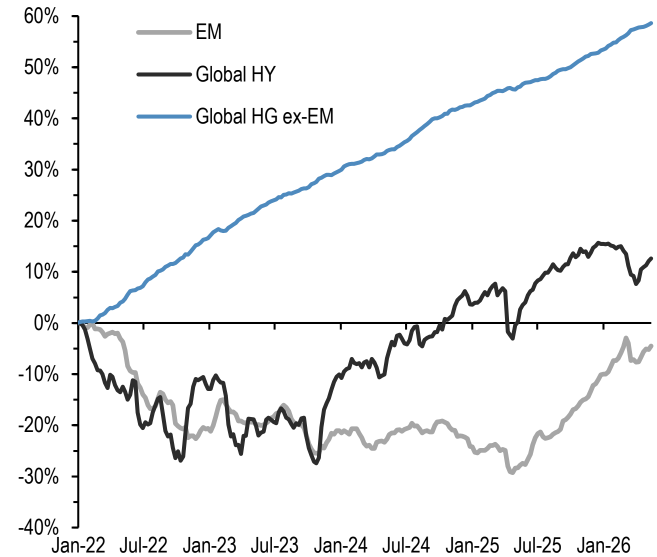
Cumulative flows into ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity. China onshore A-share ETFs have been excluded.

Chart A4: Bond ETF Flows

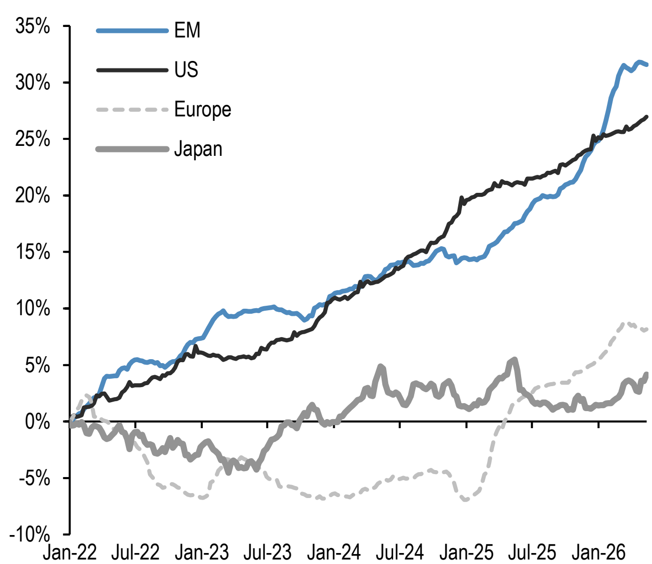
Cumulative flows into bond ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A5: Global Equity ETF Flows

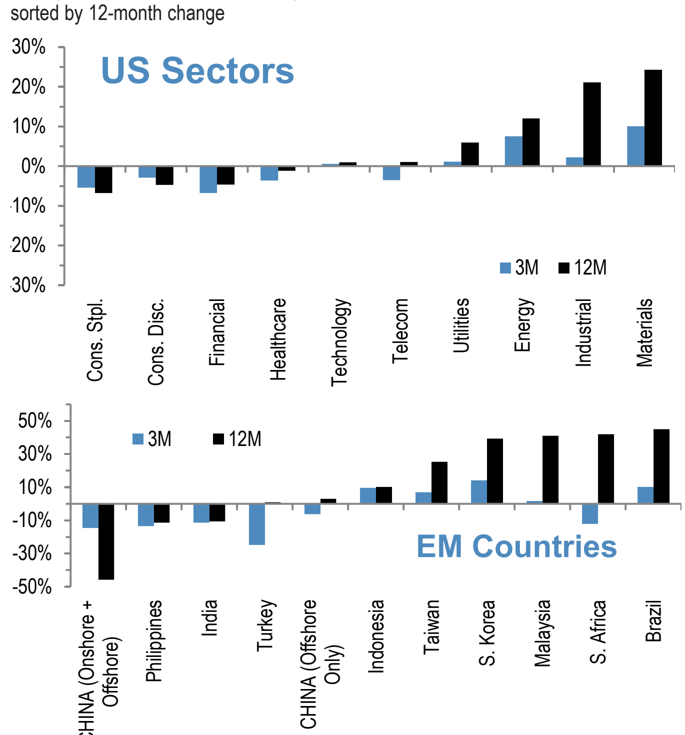
Cumulative flows into global equity ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.
Note: We include ETFs with AUM > $200mn in all the flow monitor charts. In Chart A5 for Japan, we subtract the BoJ buying of ETFs. China onshore A-share ETFs have been excluded.

Chart A6a: Equity Sectoral and Regional ETF Flows

Rolling 3-month and 12-month change in cumulative flows as a % of AUM. Both sorted by 12-month change



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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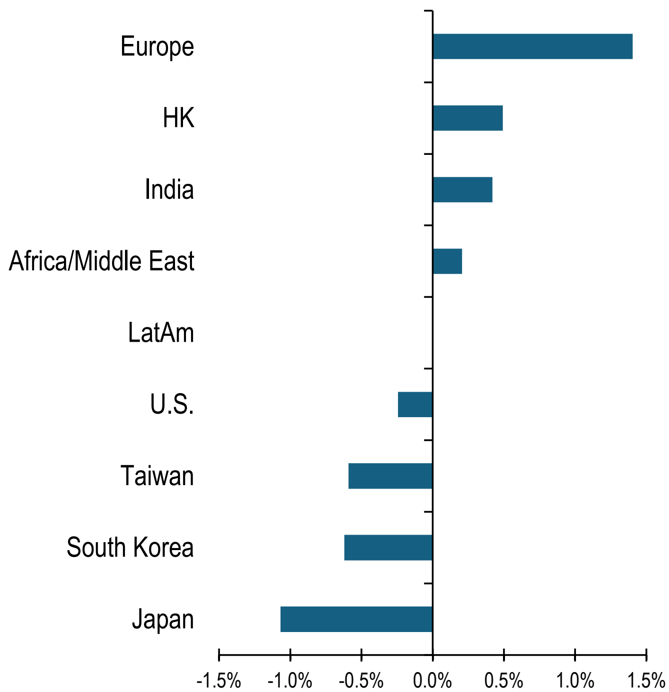
Chart A6b: Regional Equity Allocation Monitor Implied by ETFs

In %.



Source: Bloomberg Finance L.P., MSCI and J.P. Morgan Flows & Liquidity.
Note: Our regional equity allocation monitor implied by ETFs is based on a comparison of the share of each region in the equity ETF space relative to their corresponding shares in global equity indices.

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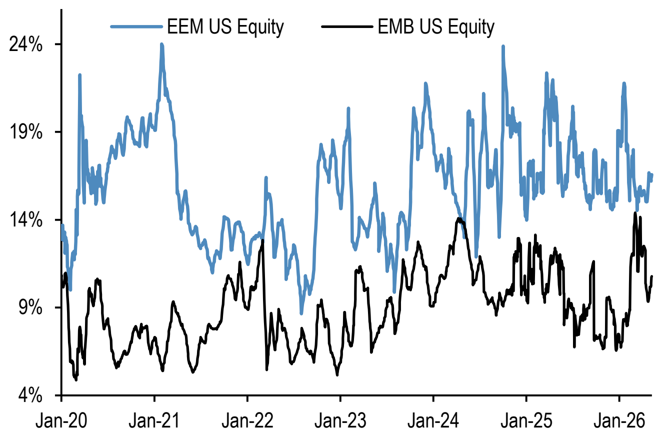
Short Interest Monitor

Chart A7: Short interest on the EEM and EMB US ETF

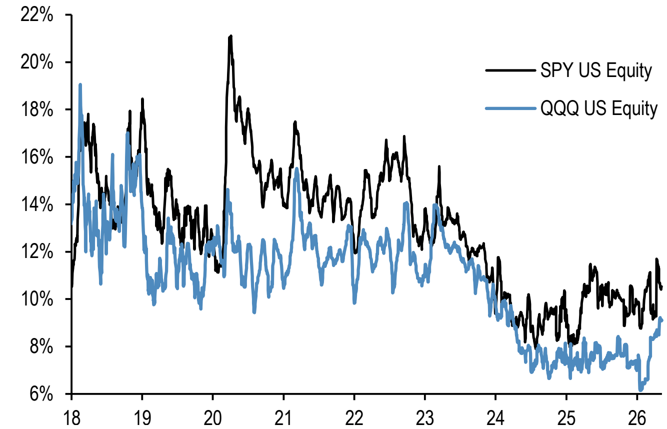
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A9: Short interest on the SPY and QQQ US ETF

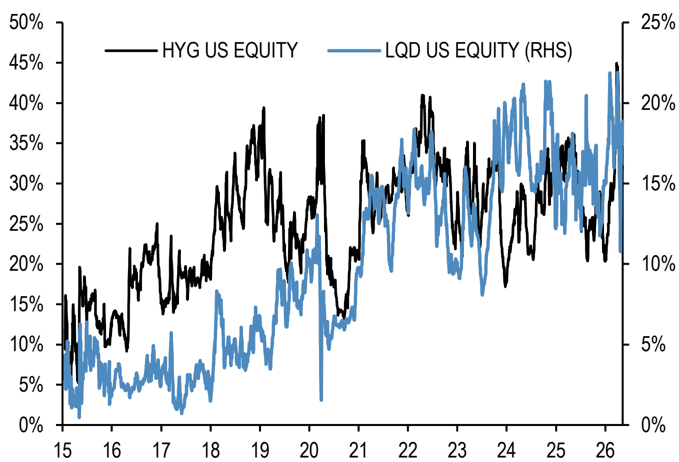
Short Interest as a % share of shares outstanding. Last obs is for 8th May 2026.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A8: Short interest on the LQD and HYG US ETF

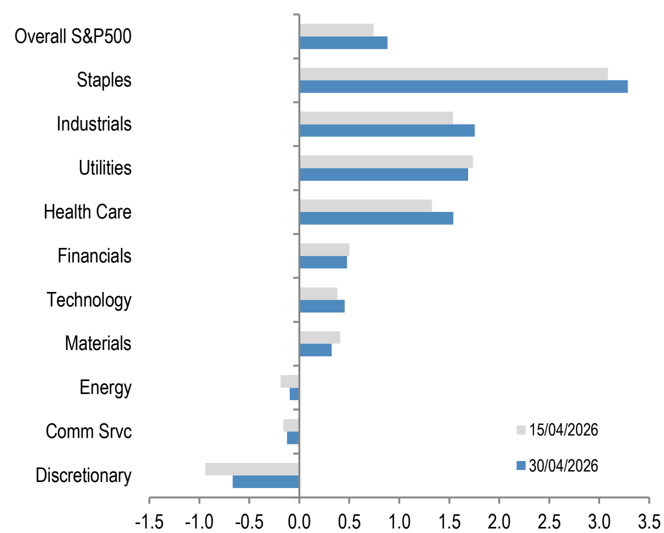
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A10: S&P500 sector short interest

Short interest as a % of shares outstanding based on z-scores. A strategy which overweights the S&P500 sectors with the highest short interest z-score (as % of shares o/s) vs. those with the lowest, produced an information ratio of 0.7 with a success rate of 56% (see F&L, Jun 28,2013 for more details).



Source: NYSE, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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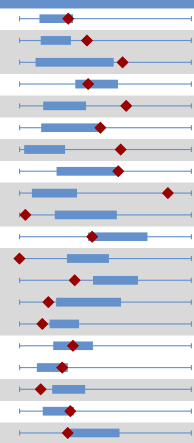
Chart A11a: Cross Asset Volatility Monitor 3m ATM Implied Volatility (1y history), as of 11th May-2026

This table shows the richness/cheapness of current three-month implied volatility levels (red dot) against their one-year historical range (thin blue bar) and the ratio to current realised volatility. Assets with implied volatility outside their 25th/75th percentile range (thick blue bar) are highlighted. The implied-to-realised volatility ratio uses 3-month implied volatilities and 1-month (around 21 trading days) realised volatilities for each asset.

Asset	Current	Low	Low date	High	High date	Upside	Downside	Implied/realized volatility
S&P 500	16%	13%	28-Aug-25	24%	27-Mar-26	8%	3%	1.55x
EuroSTOXX	18%	13%	18-Sep-25	26%	27-Mar-26	8%	5%	0.95x
Nikkei 225	27%	18%	22-Jul-25	33%	31-Mar-26	6%	9%	1.03x
Hang Seng	20%	17%	23-Dec-25	24%	26-Mar-26	4%	3%	1.20x
MSCI EM	26%	12%	15-Aug-25	35%	27-Mar-26	9%	14%	1.10x
Gold	23%	14%	19-Aug-25	33%	26-Mar-26	10%	9%	1.09x
Oil (brent)	63%	27%	19-Sep-25	89%	30-Mar-26	26%	36%	0.85x
Copper	27%	14%	18-Aug-25	37%	03-Feb-26	10%	13%	1.08x
BB commodity index	23%	13%	10-Dec-25	25%	30-Mar-26	2%	10%	1.26x
EUR/USD	9%	9%	03-Oct-25	13%	26-May-25	4%	0%	1.78x
USD/NOK	9%	8%	17-Dec-25	11%	30-Mar-26	2%	1%	1.68x
USD/JPY	8%	8%	28-Apr-26	12%	23-May-25	4%	0%	0.80x
GBP/USD	7%	6%	12-Dec-25	8%	02-Jul-25	2%	1%	1.22x
USD/CHF	7%	6%	09-Jan-26	10%	21-May-25	3%	1%	0.95x
10y US swaps	7%	6%	13-May-25	10%	27-Mar-26	3%	1%	1.03x
10y Eur swaps	59%	43%	26-Feb-26	97%	27-Mar-26	37%	17%	0.91x
CDX IG	45%	38%	29-Jul-25	66%	27-Mar-26	21%	7%	1.00x
CDX HY	34%	31%	09-Jan-26	57%	30-Mar-26	23%	3%	0.89x
iTraxx	47%	37%	09-Jan-26	70%	27-Mar-26	23%	9%	0.90x
iTraxx X/O	21%	19%	23-Dec-25	26%	17-Oct-25	5%	2%	0.42x



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Note: Swaps volatility is 3m 10y payer ATMF implied annualized BP vol and credit volatility is 3m 5y on-the-run ATM spread volatility. MSCI EM, Gold, Oil, Copper, BB Commodity Index and Treasury futures are 3m implied vol from Bloomberg.

Definitions:

Current:	Latest available closing level (11-May-26)
Low:	Lowest closing level in the last 1y
Low date:	Date the lowest closing level was reached (or the first time it was reached in the case of several identical low closing levels)
High:	Highest closing level in the last 1y
High date:	Date the highest closing level was reached (or the first time it was reached in the case of several identical high closing levels)
Graph:	Shows the current level and the 25th/75th percentile relative to the 1y high/low
Upside:	Implied return/volatility percentage points from current level up to the High (note: return is calculated as simple difference for spread products)
Upside (σ):	Upside in terms of standard deviations (Upside / Current 1y realized volatility)
Downside:	Implied return/volatility percentage points from current level down to the Low (note: return calculated as simple difference for spread products)
Downside (σ):	Downside in terms of standard deviations (Downside / Current 1y realized volatility)
Implied/realized volatility:	Current 3m implied volatility / current realized 3m volatility

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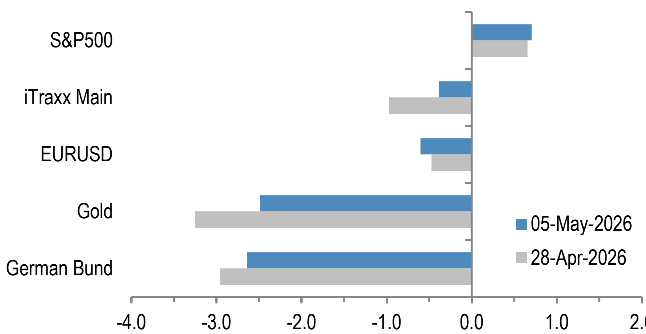
Chart A11b: Option skew monitor

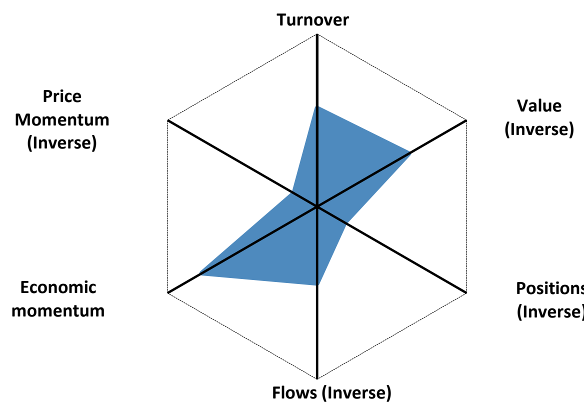
Skew is the difference between the implied volatility of out-of-the-money (OTM) call options and put options. A positive skew implies more demand for calls than puts and a negative skew, higher demand for puts than calls. It can therefore be seen as an indicator of risk perception in that a highly negative skew in equities is indicative of a bearish view. The chart shows z-score of the skew, i.e. the skew minus a rolling 2-year avg skew divided by a rolling two-year standard deviation of the skew. A negative skew on iTraxx Main means investors favour buying protection, i.e. a short risk position. A positive skew for the Bund reflects a long duration view, also a short risk position.



Source: J.P. Morgan Flows & Liquidity.

Chart A11c: Equity market health map

Each of the six axes corresponds to a key indicator for markets. The position of the blue line on each axis shows how far the current observation is from the extremes at either end of the scale. Overall, the larger the blue area within the hexagon, the better the outlook for the S&P500. All variables are expressed as the percentile of the distribution that the observation falls into. I.e. a reading in the middle of the axis means that the observation falls exactly at the median of all historical observations. **Turnover**: Percentile rank of the 21-day rolling S&P500 turnover since 2007. **Value**: Inverse of percentile rank of our quarterly S&P 500 Equity Risk Premium (ERP) estimate since 1957 as described in "A fair value model for US Bonds, Credit and Equities", Panigirtzoglou and Loeys, Jun 2005. **Positions**: Percentile rank of the difference between government bonds and equity positioning from the cross-asset positioning monitor on the front page of our publication since 2016. **Flows**: Inverse of percentile rank of the difference between equity and bond fund flows from the cross-asset fund flow monitor on the front page of our publication since 2007. **Economic momentum**: Percentile rank of the 2-month change in the global manufacturing PMI since 1998, see "REVISITING: Using the Global PMI as trading signal", Nikolaos Panigirtzoglou, Jan 2012). **Price momentum**: Inverse of the percentile rank of the z-score of the difference in the momentum signal of S&P500 vs. 10y USTs since 2016 (based on the average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 in the Appendix). China onshore A-share ETFs have been excluded. Last obs. as of 8th May.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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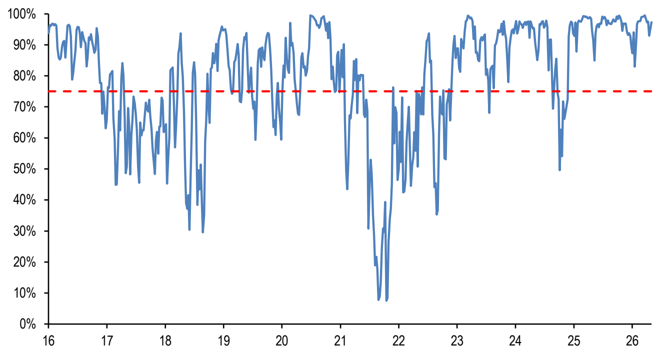
Chart A11d: Implied probability of the S&P500 rising over the next six months with a cutoff set at 75%



Source: J.P. Morgan Flows & Liquidity.

Chart A11e: Logistic regression model for direction of the S&P 500 over a 6 month horizon & Ex-ante probability of an up-move from the logistic regression algorithm vs. the ex-post realised 6-month return in the S&P 500

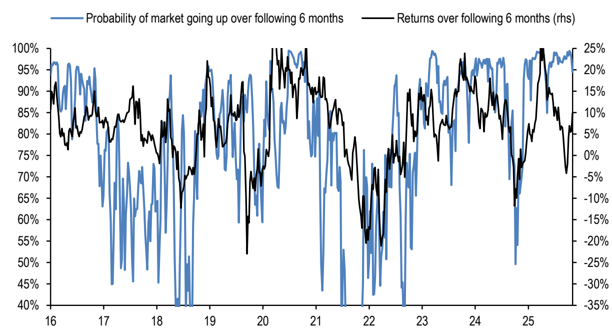
In %.

Dependent Variable: S&500 UP (1) or DOWN (0) over the next six months
Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)

Sample (adjusted): 1/08/2016 12/20/2024
Included observations: 468 after adjustments

Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	3.42	0.84	4.08	0.00
TURNOVER	2.97	0.72	4.15	0.00
VALUE	-3.47	0.90	-3.88	0.00
POSITIONS	-0.77	0.55	-1.39	0.16
FLOWS	-1.79	0.53	-3.38	0.00
ECON_MOMENTUM	3.35	0.62	5.42	0.00
MOMENTUM	-1.36	0.62	-2.19	0.03



Source: J.P. Morgan Flows & Liquidity.

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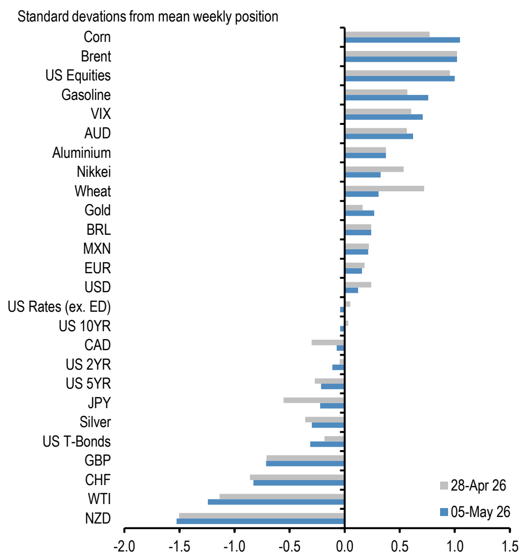
Spec position monitor

Chart A12: Weekly Spec Position Monitor

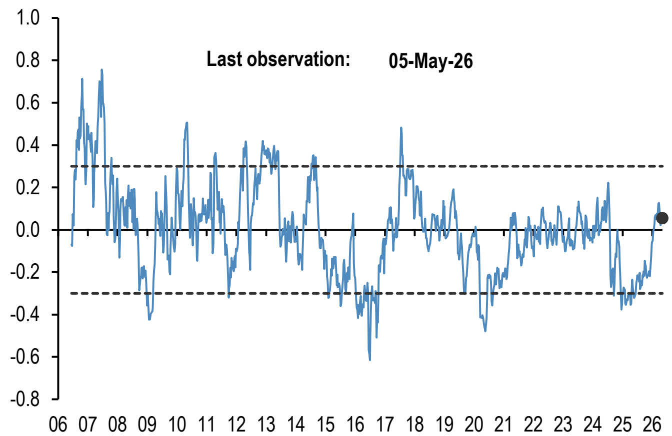
Net spec positions are proxied by the number of long contracts minus the number of short contracts using the speculative category of the Commitments of Traders reports (as reported by CFTC). To proxy for speculative investors for equity and US Treasury bond futures positions we use Asset managers and leveraged funds (see Chart A13), whereas for other assets we use the legacy Non-Commercial category. This net position is then converted to a dollar amount by multiplying by the contract size and then the corresponding futures price. We then scale the net positions by open interest. The chart shows the z-score of these net positions. US rates is a duration-weighted composite of the individual UST futures contracts excluding the Eurodollar contract.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A14: Spec position indicator on Risky vs. Safe currencies

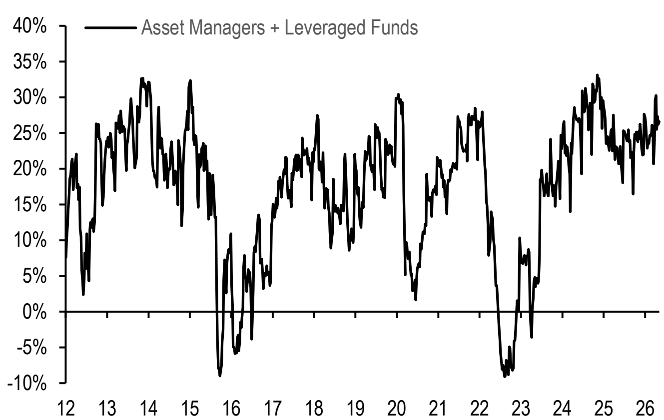
Difference between net spec positions on risky & safe currencies. Net spec position is calculated in USD across 5 'risky' and 3 'safe' currencies (safe currencies also include Gold). These positions are then scaled by open interest and we take an average of 'risky' and 'safe' assets to create two series. The chart is then simply the difference between the"risky" and "safe" series. The final series shown in the chart below is demeaned using data since 2006. The risky currencies are: AUD, NZD,CAD, MXN and BRL. The safe currencies are: JPY, CHF and Gold.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A13: Positions in US equity futures by Asset managers and Leveraged funds

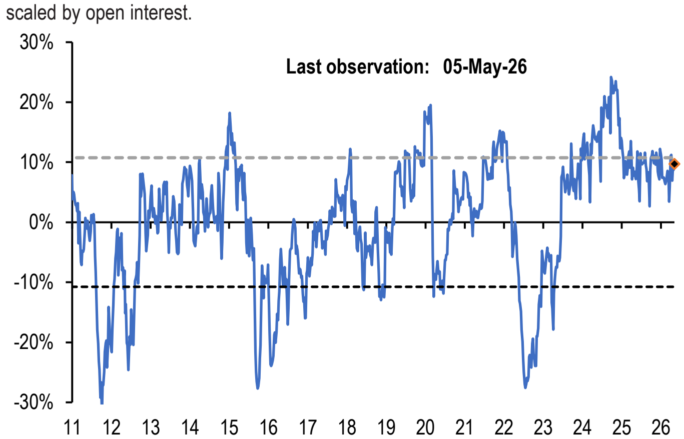
CFTC positions in US equity futures by Leveraged funds and Asset managers (as a % of open interest). It is an aggregate of the S&P500, DowJones, NASDAQ and their Mini futures contracts.



Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

Chart A15: Spec position indicator on US equity futures vs. intermediate sector UST futures

Difference between net spec positions on US equity futures vs. intermediate sector UST futures. This indicator is derived by the difference between total CFTC positions in US equity futures by Asset managers + Leveraged Funds scaled by open interest minus the Asset managers + Leveraged Funds spec position on intermediate sector UST futures (i.e. all UST futures duration weighted ex ED and ex 2Y UST futures) also scaled by open interest.



Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

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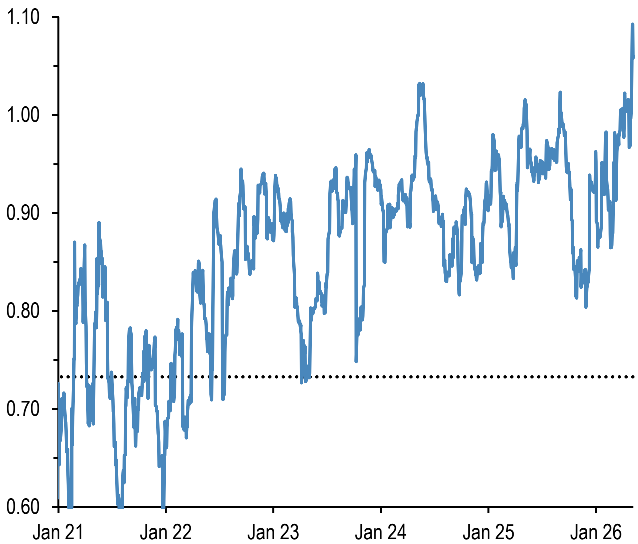
Mutual fund and hedge fund betas

Chart A16: 21-day rolling beta of 20 biggest active US bond mutual fund managers with respect to the US Agg Bond Index

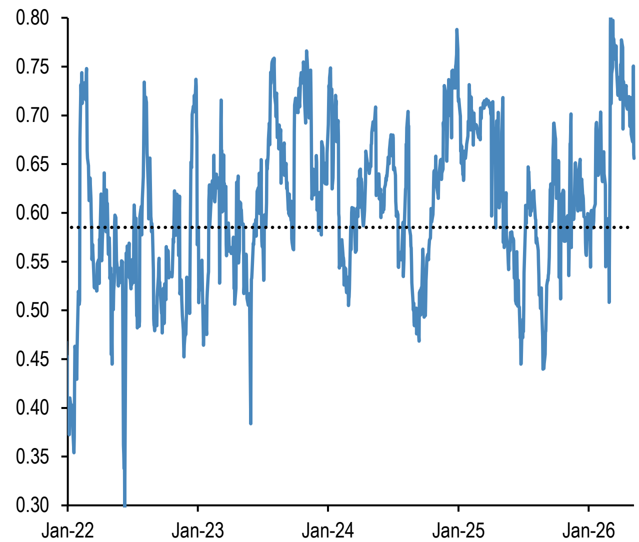
The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A17: 21-day rolling beta of 20 biggest active Euro bond mutual fund managers with respect to the Euro Agg Bond Index

The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A18: Performance of various types of investors

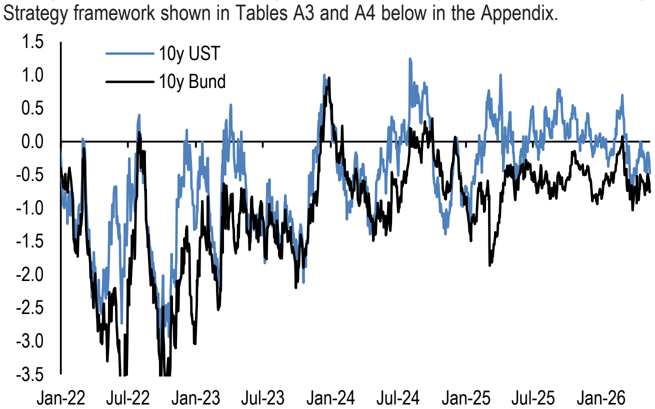
The table depicts the performance of various types of investors in % as of 8th May 2025.

Date	2020	2021	2022	2023	2024	2025	2026
Investors							
Equity L/S	13.4%	9.3%	-7.0%	11.3%	13.0%	15.5%	6.3%
Macro ex-CTAs	4.9%	4.3%	11.5%	1.3%	6.0%	10.2%	3.7%
CTAs	1.2%	8.8%	14.7%	-2.5%	2.9%	-1.1%	8.7%
Risk Parity Funds	3.5%	4.7%	-18.6%	6.0%	5.0%	17.3%	6.9%
US Balanced MFs	13.2%	14.4%	-13.0%	13.8%	11.4%	14.4%	5.2%
Benchmark							
MSCI AC World	16.3%	16.4%	-18.4%	22.2%	17.5%	22.3%	9.5%
Barclays Global Agg	5.6%	-2.5%	-11.2%	7.1%	3.4%	4.9%	0.5%
S&P Riskparity Vol 10	11.5%	12.8%	-16.2%	15.0%	5.4%	15.9%	14.8%
60 US Equity : 40 US Bonds	13.3%	14.8%	-15.4%	18.6%	16.4%	12.7%	5.5%

Source: Bloomberg Finance L.P., HFR, Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A19: Momentum signals for 10Y UST and 10Y Bunds

Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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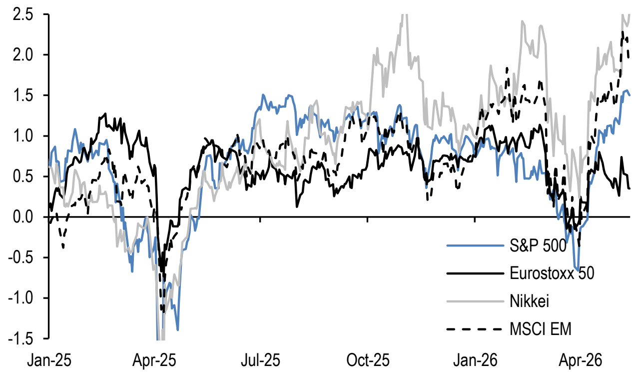
Chart A20: Momentum signals for S&P500

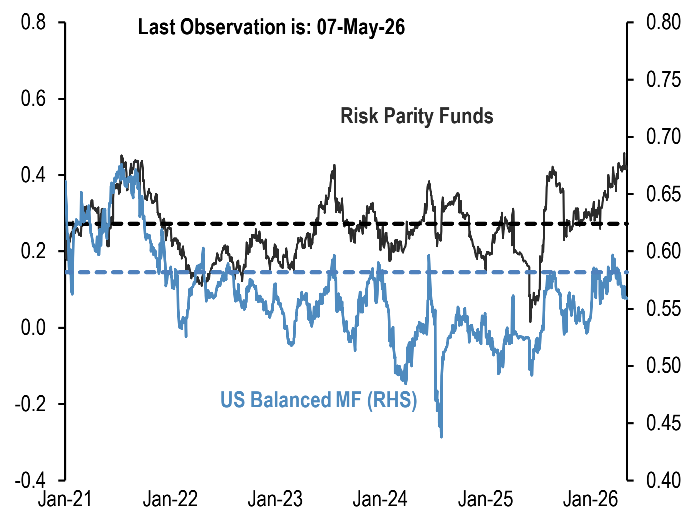
Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A21: Equity beta of US Balanced Mutual funds and Risk Parity funds

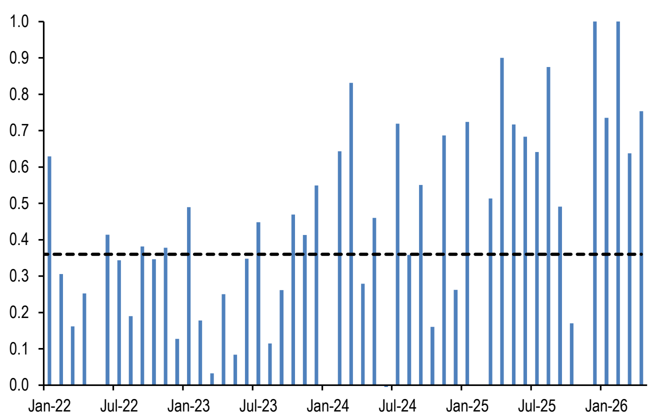
Rolling 42-day equity beta based on a bivariate regression of the daily returns of our Balanced Mutual fund and Risk Parity fund return indices to the daily returns of the S&P 500 and BarCap US Agg indices. Given that these funds invest in both equities and bonds we believe that the bivariate regression will be more suitable for these funds. Our risk parity index consists of 40 daily reporting Risk Parity funds. Our Balanced Mutual fund index includes the top 20 US-based active funds by assets and that have existed since 2006. Our Balanced Mutual fund index has a total AUM of $700bn, which is around half of the total AUM of $1.5tr of US based Balanced funds which we believe to be a good proxy of the overall industry It excludes tracker funds and funds with a low tracking error. Dotted lines are average since 2015.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A22: Equity beta of monthly reporting Equity Long/Short hedge funds

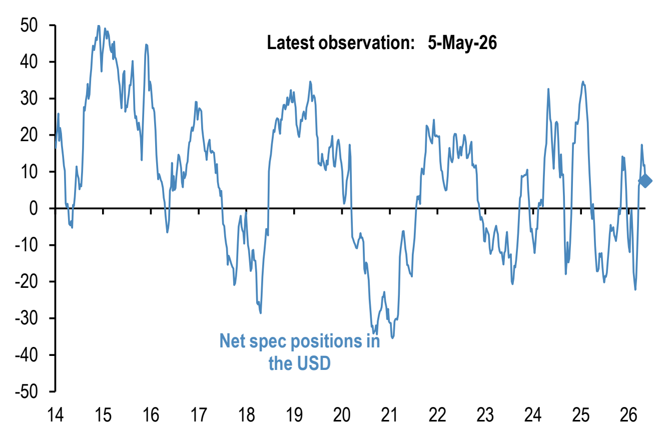
Proxied by the ratio of the monthly performance of Pivotal Path Asset-Weighted Equity Diversified hedge fund index divided by the monthly performance of MSCI AC World Index. Latest obs. is Apr 26 based on provisional data.



Source: Bloomberg Finance L.P., Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A23: USD exposure of currency hedge funds

The net spec position in the USD as reported by the CFTC. Spec is the non-commercial category from the CFTC.



Source: CFTC, Barclays, Datastream, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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CTAs – Trend following investors' momentum indicators

Table A3: Simple return momentum trading rules across various commodities

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg, days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
WTI	short	21	5.7	22.7	0.25	1	2	1.1	6.7%
	long	462				0	16	2.4	69.2%
Brent	short	84	4.4	21.5	0.20	0	51	2.6	31.2%
	long	504				0	16	2.5	71.8%
Unleaded gas	short	105	2.6	23.2	0.11	0	49	3.4	46.3%
	long	483				0	43	2.9	77.5%
Heat Oil	short	63	4.8	21.9	0.22	0	2	2.1	22.4%
	long	483				0	50	3.5	106.6%
Gasoil	short	63	8.5	21.1	0.40	0	51	2.3	24.4%
	long	378				0	50	3.8	108.0%
Nat gas	short	105	13.4	36.0	0.37	-1	42	-0.5	-9.4%
	long	315				-1	71	-0.8	-23.6%
EU emission allowances	short	42	9.9	28.9	0.34	1	23	0.4	4.2%
	long	483				0	0	0.0	0.7%
1m fwd TTF Nat gas	short	21	36.4	31.3	1.16	1	1	0.6	5.5%
	long	294				1	36	0.8	35.6%
Gold	short	21	3.0	10.6	0.29	-1	0	-0.2	-0.6%
	long	483				0	311	2.2	29.2%
Silver	short	10	6.4	19.7	0.33	0	3	3.0	10.3%
	long	462				0	163	3.2	77.9%
Palladium	short	42	12.0	23.0	0.52	-1	0	-0.2	-1.5%
	long	273				1	32	0.2	4.3%
Platinum	short	105	4.3	18.6	0.23	-1	16	-0.2	-2.3%
	long	273				1	0	1.5	23.1%
Aluminium	short	105	5.8	15.2	0.38	1	4	1.4	10.4%
	long	357				0	31	1.9	27.6%
Copper	short	147	7.5	17.2	0.43	1	63	1.1	13.3%
	long	399				1	69	1.4	31.5%
Lead	short	126	0.7	19.3	0.03	0	0	0.0	-0.1%
	long	357				-1	82	-0.2	-3.9%
Nickel	short	42	12.0	23.0	0.52	1	21	0.6	4.9%
	long	336				1	95	0.6	14.7%
Zinc	short	126	8.0	19.6	0.41	1	31	0.8	8.6%
	long	399				1	161	0.8	18.5%
Wheat	short	168	4.5	23.2	0.19	0	0	1.6	19.3%
	long	294				1	21	1.1	16.7%
Kansas wheat	short	147	8.6	20.8	0.41	0	0	2.0	24.4%
	long	483				1	18	0.8	15.9%
Corn	short	63	7.7	16.5	0.46	1	16	0.4	3.4%
	long	399				-1	5	-0.1	-1.2%
Soybeans	short	42	5.1	14.5	0.35	1	9	0.6	3.3%
	long	231				1	69	0.7	9.1%
Cotton	short	168	6.2	18.5	0.33	0	1	1.6	22.3%
	long	483				1	19	0.6	13.9%
Sugar	short	63	6.2	21.6	0.29	1	2	0.3	3.2%
	long	252				-1	31	-0.2	-3.1%
Coffee	short	63	7.9	23.8	0.33	-1	4	-0.3	-3.0%
	long	273				-1	72	-0.6	-12.2%
Cocoa*		10	4.9	30.9	0.16	0	6	4.0	14.8%

* For cocoa, uses only short-term momentum and a z-score threshold of 3 rather than 1.5 as for other contracts.

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Table A4: Simple return momentum trading rules across international equity indices, bond futures and FX

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg, days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
S&P 500	short	84	7.6	12.3	0.62	0	2	1.5	7.6%
	long	315				0	2	1.5	15.7%
Nasdaq 100	short	84	9.9	16.2	0.61	0	4	1.8	14.2%
	long	462				1	0	1.5	29.4%
Nikkei	short	63	4.0	13.3	0.30	0	3	1.9	11.1%
	long	420				0	29	2.9	43.2%
FTSE 100	short	168	5.2	12.1	0.43	1	48	0.6	3.6%
	long	504				0	31	1.5	17.4%
Eurostoxx 50	short	168	4.7	16.6	0.28	1	24	0.2	1.8%
	long	294				1	31	0.5	6.0%
MSCI EM	short	42	12.9	11.5	1.12	0	6	1.9	9.8%
	long	357				0	20	2.0	31.8%
HYG credit	short	168	0.4	3.4	0.12	1	54	0.1	0.1%
	long	273				1	1	0.2	0.5%
LQD credit	short	10	2.6	6.5	0.41	0	5	0.0	0.0%
	long	189				-1	1	-0.2	-0.8%
2Y USTs	short	252	0.8	1.1	0.78	-1	51	-0.8	-0.9%
	long	462				-1	49	-0.5	-1.0%
5Y USTs	short	252	1.5	2.8	0.54	-1	44	-0.6	-1.5%
	long	420				-1	15	-0.3	-1.1%
10Y USTs	short	42	2.0	3.6	0.56	-1	15	-0.6	-0.8%
	long	441				-1	15	-0.3	-1.3%
2Y Schatz	short	189	0.6	0.8	0.82	-1	20	-1.3	-1.1%
	long	504				0	1	-1.5	-2.7%
5y Bobl	short	63	1.4	1.9	0.73	-1	24	-0.9	-0.9%
	long	483				-1	301	-1.0	-3.5%
10y Bund	short	105	2.2	3.4	0.65	-1	48	-0.7	-1.4%
	long	462				-1	650	-0.9	-4.4%
10Y JGB	short	126	0.9	2.3	0.37	0	1	-1.7	-1.8%
	long	273				0	48	-2.5	-3.6%
10Y Gilts	short	42	2.3	4.0	0.58	-1	4	-1.4	-2.3%
	long	399				-1	177	-1.3	-6.9%
10y OAT vs Bund	short	189	0.4	1.5	0.28	1	31	0.8	0.7%
	long	294				1	52	0.9	0.9%
10y BTP vs. Bund	short	84	2.3	6.3	0.36	0	1	0.0	0.0%
	long	273				1	8	0.2	0.7%
Euro	short	42	3.3	6.4	0.51	1	8	0.1	0.3%
	long	273				-1	51	-0.4	-2.1%
Yen	short	21	3.5	6.3	0.56	1	8	0.1	0.2%
	long	462				-1	8	-1.3	-10.8%
Sterling	short	168	1.3	7.1	0.19	-1	0	-0.1	-0.3%
	long	294				-1	53	-0.2	-1.3%
AUD	short	42	3.9	7.8	0.50	1	23	0.7	1.9%
	long	420				1	78	0.7	6.2%
CAD	short	168	1.3	5.9	0.22	-1	6	-0.3	-1.0%
	long	504				-1	288	-0.7	-4.4%
G10 FX	short	42	2.1	5.2	0.41	-1	21	-0.5	-0.7%
	long	189				-1	4	-0.1	-0.2%
EMCI	short	42	2.6	5.0	0.53	1	16	1.0	1.8%
	long	189				1	217	1.1	4.2%

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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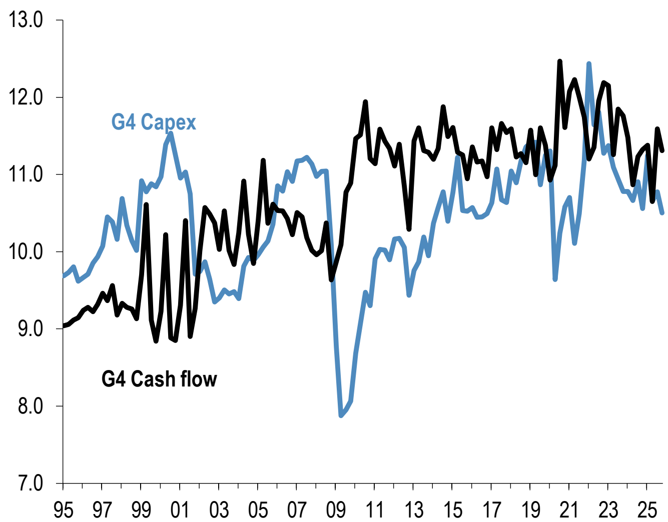
Corporate Activity

Chart A24: G4 non-financial corporate capex and cash flow as % of GDP

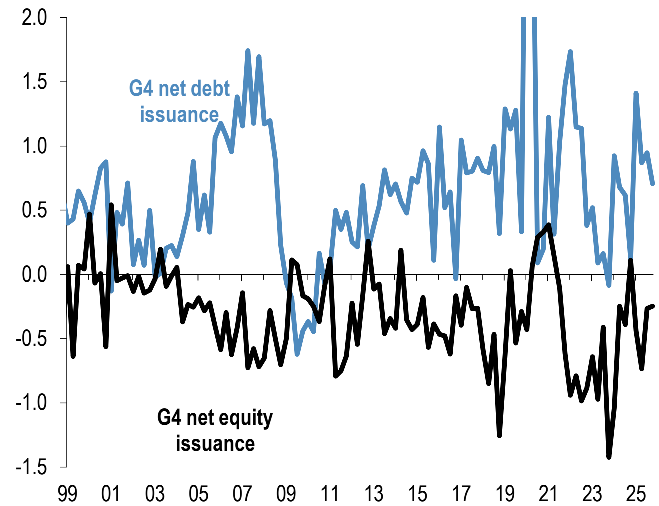
% of GDP, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A25: G4 non-financial corporate sector net debt and equity issuance

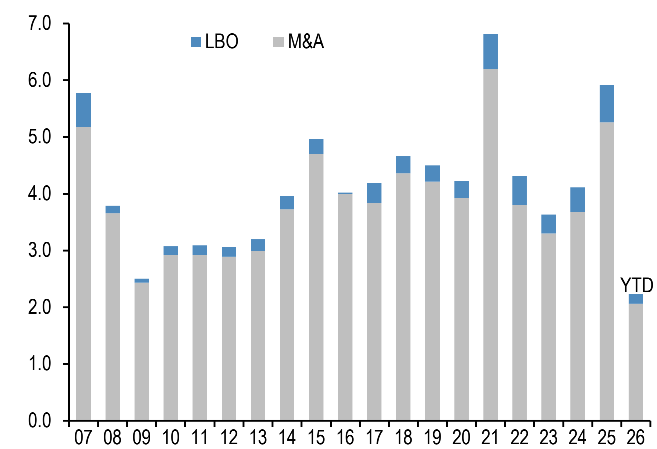
$tr per quarter, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A26: Global M&A and LBO

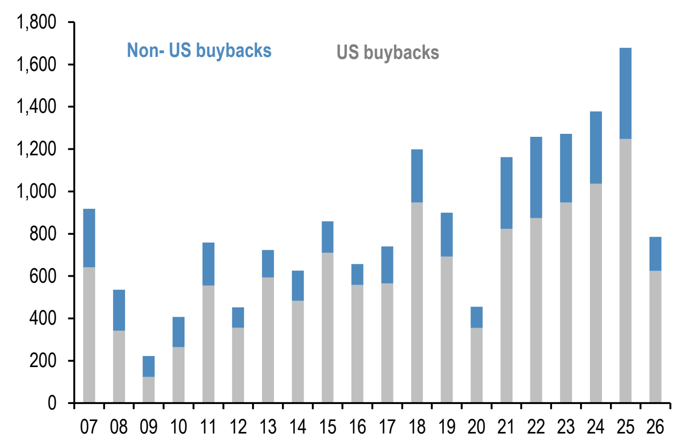
$tr. M&A and LBOs are announced.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Chart A27: US and non-US share buyback

$bn, are as of May 26. Buybacks are as announced.



Source: Bloomberg Finance L.P., LSEG, J.P. Morgan Flows & Liquidity.

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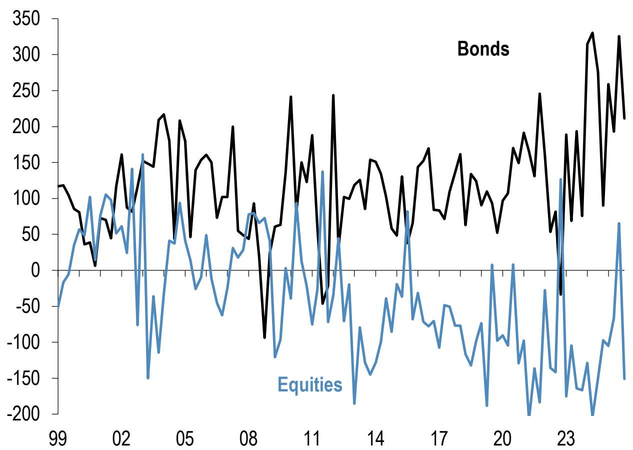
Pension fund and insurance company flows

Chart A28: G4 pension funds and insurance companies equity and bond flows

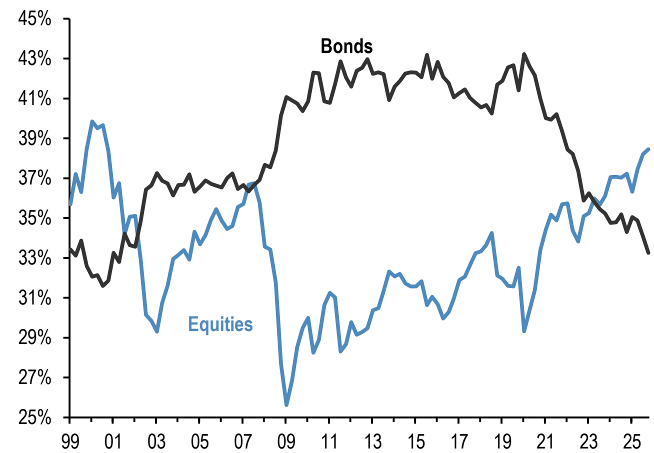
Equity and bond buying in $bn per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A29: G4 pension funds and insurance companies equity and bond levels

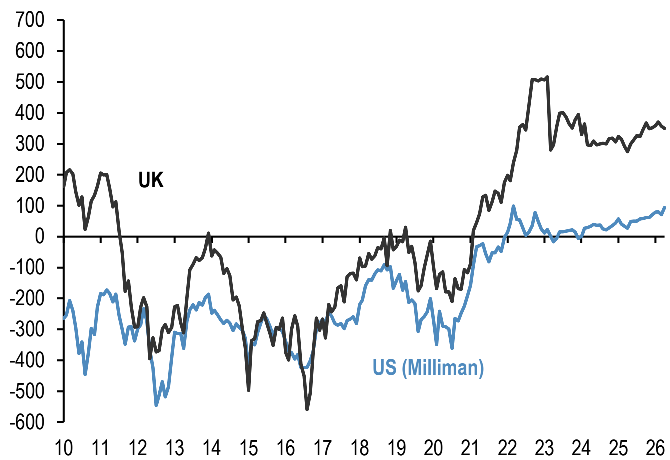
Equity and bond as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A30: Pension fund deficits

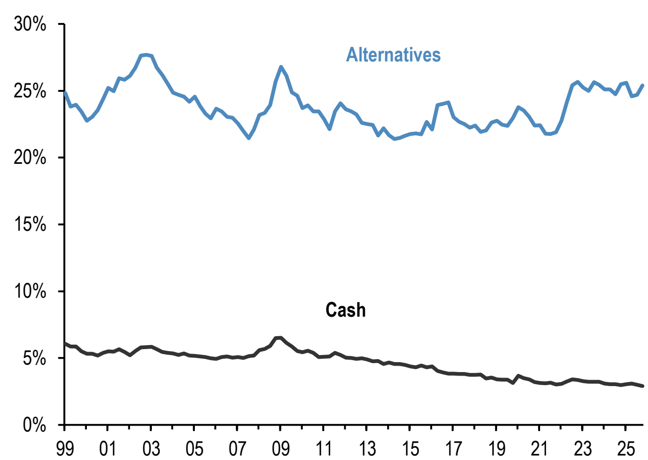
US$bn. For US, funded status of the 100 largest corporate defined benefit pension plans, from Milliman. For UK, funded status of the defined benefit schemes eligible for entry to the Pension Protection Fund, converted to US$ at today's exchange rates. Last obs. is Apr'26 for US & UK.



Source: Milliman, UK Pension Protection Fund, J.P. Morgan Flows & Liquidity.

Chart A31: G4 pension funds and insurance companies cash and alternatives levels

Cash and alternative investments as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

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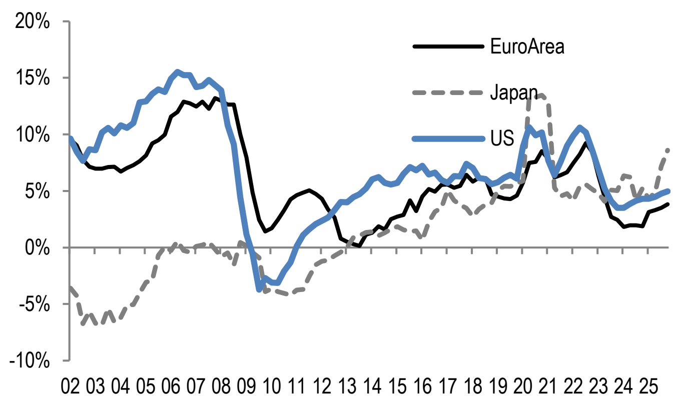
Credit Creation

Chart A32: Credit creation in the US, Japan and Euro area

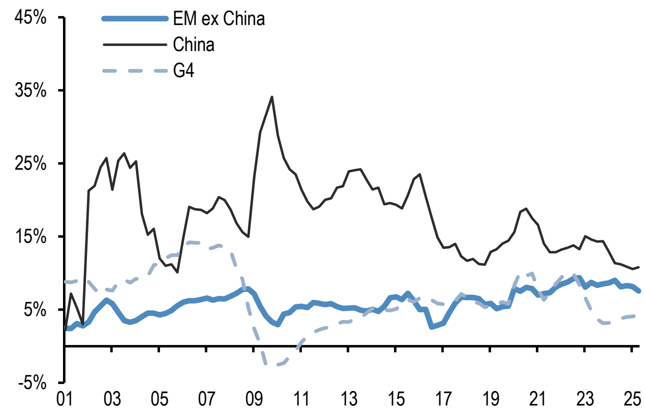
Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. Last obs. is Q4'25 for Japan, Euro Area and US.



Source: Fed, ECB, BoJ, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A33: Credit creation in EM

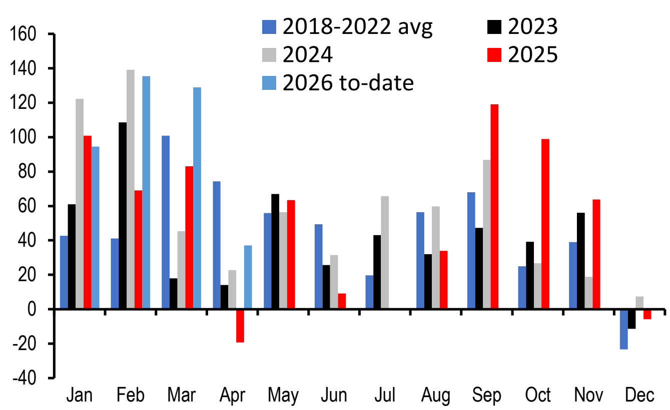
Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. The chart uses the FX rates as of the latest observation to calculate the credit creation at constant exchange rates. Last obs. is for Q2'25.



Source: G4 Central banks FoF, BIS, ICI, Barcap, Bloomberg Finance L.P., IMF, and J.P. Morgan Flows & Liquidity.

Chart A34: Monthly net issuance of US HG bonds

$bn. Up to Apr 23rd.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Table A5: Equity and Bond issuance

$bn, Equity supply and corporate announcements are based on announced deals, not completed. M&A is announced deal value and buybacks are announced transactions. Y/Y change is change in YTD announcements over the same period last year.

Equity Supply	1-May	4 wk avg	13 wk avg	y/y chng
Global IPOs	5.9	4.9	3.7	72%
Secondary Offerings	10.0	9.1	10.4	21%
Corporate announcements				
M&A - Global	141.5	114.2	126.5	31%
- US Target	48.6	52.7	58.8	51%
- Non-US Target	92.9	61.5	67.7	17%
Net bond issuance	**Feb-26**	**3 mth avg**	**YTD avg**	**y/y chng**
USD	8	144	179	-50%
Non-USD	162	129	277	-4%

Source: Bloomberg Finance L.P., Dealogic, Thomson Reuters, J.P. Morgan Flows & Liquidity.

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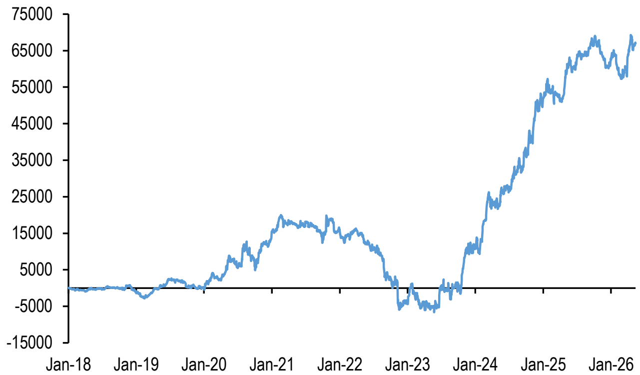
Bitcoin monitor

Chart A35: Our Bitcoin position proxy, based on open interest in CME Bitcoin futures contracts

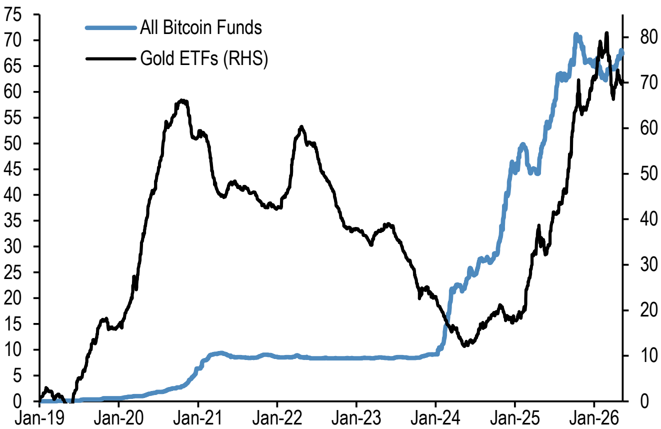
In number of contracts. Last obs. for 12th May2026.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A36: Cumulative Flows in all Bitcoin funds and Gold ETF holdings

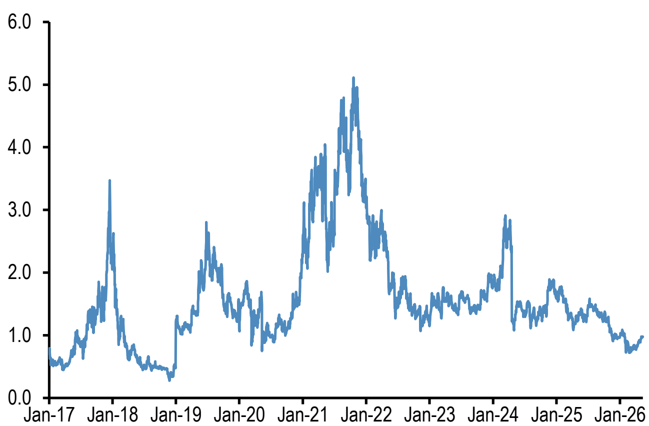
Both the y-axis in $bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A37: Ratio of bitcoin market price to production cost

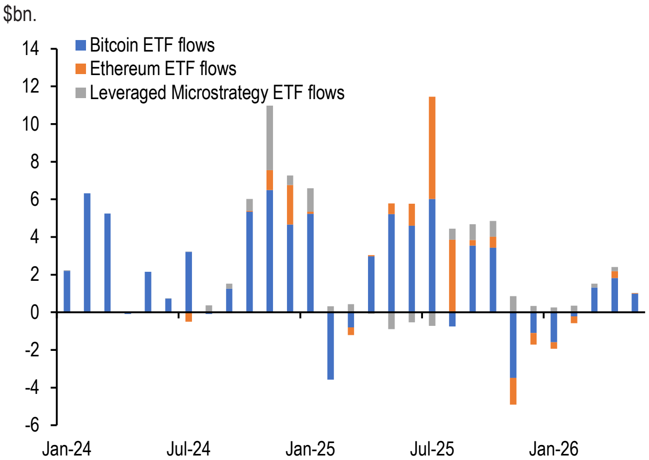
Based on the cost of production approach following Hayes (2018).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A38: Monthly flows into spot bitcoin ETFs, spot ethereum ETFs and leveraged MicroStrategy ETF

$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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Japanese flows and positions

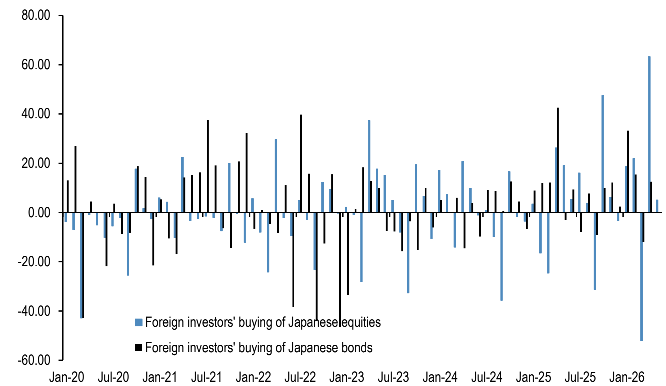
Chart A39: Tokyo Stock Exchange margin trading: total buys minus total sells

In bn of shares. Topix on right axis.

Source: Tokyo Stock Exchange, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A40: Monthly net purchases of Japanese bonds and Japanese equities by foreign residents

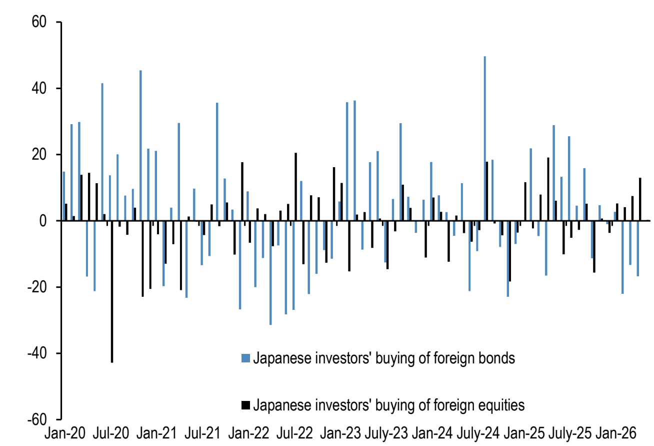
$bn, Last weekly obs. is for 1st May' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A41: Monthly net purchases of foreign bonds and foreign equities by Japanese residents

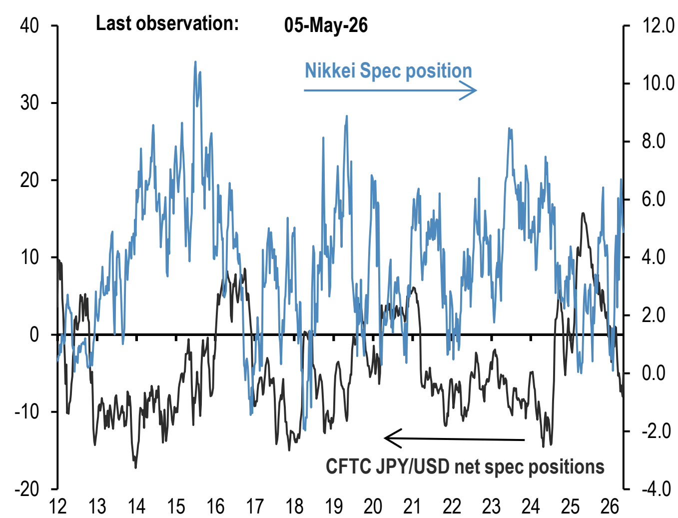
$bn, Last weekly obs. is for 1st May' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A42: Overseas CFTC spec positions

CFTC spec positions are in $bn. For Nikkei we use CFTC positions in Nikkei futures (USD & JPY) by Leveraged funds and Asset managers.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

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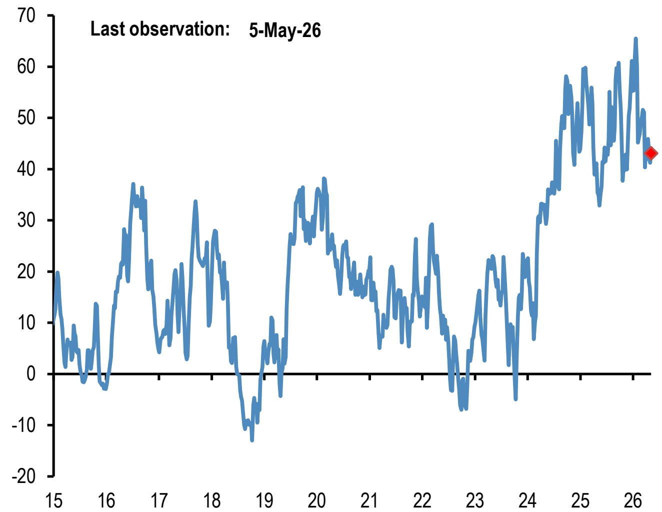
Commodity flows and positions

Chart A43: Gold spec positions

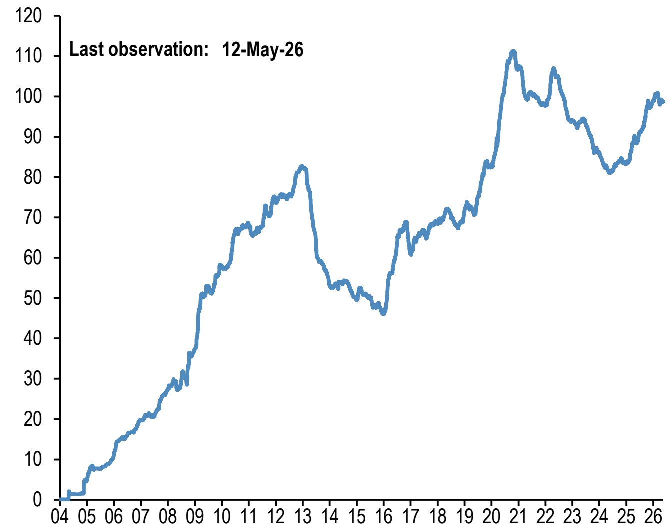
$bn. CFTC net long minus short position in futures for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A44: Gold ETFs

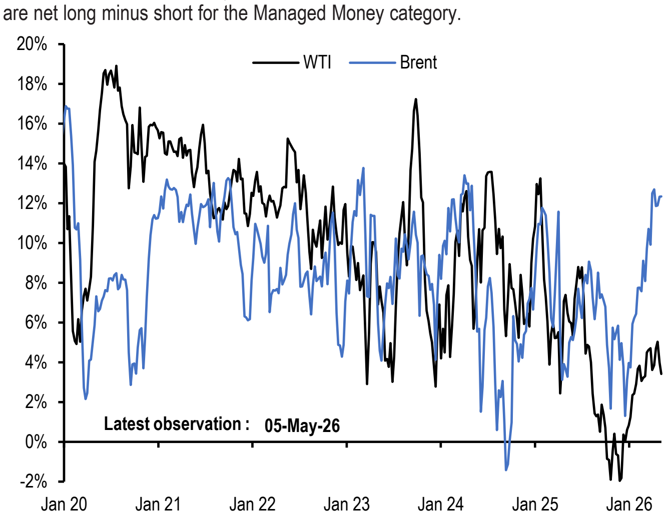
Mn troy oz. Physical gold held by all gold ETFs globally.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A45: Oil spec positions

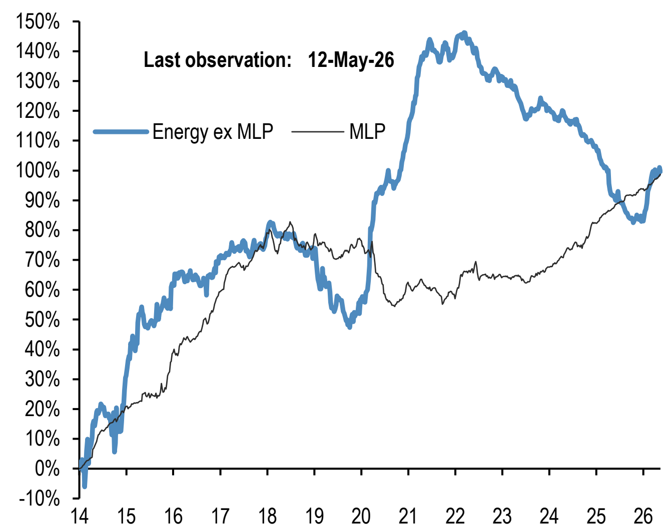
Net spec positions divided by open interest. CFTC futures positions for WTI and Brent are net long minus short for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A46: Energy ETF flows

Cumulative energy ETFs flows as a % of AUM. MLP refers to the Alerian MLP ETF.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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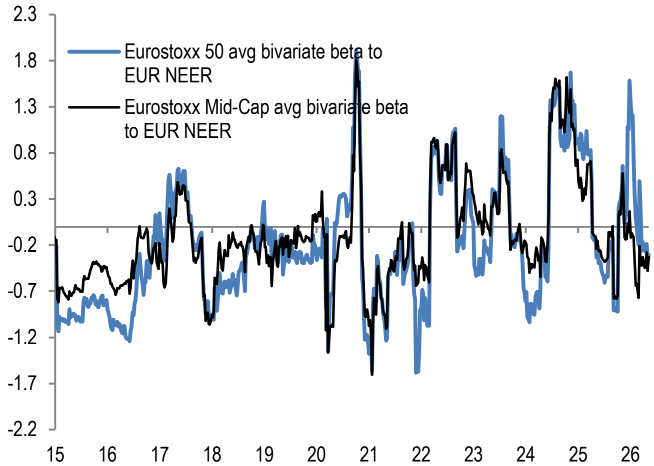
Corporate FX hedging proxies

Chart A47: Average beta of Eurostoxx 50 companies and Eurostoxx Mid-Cap to trade-weighted EUR

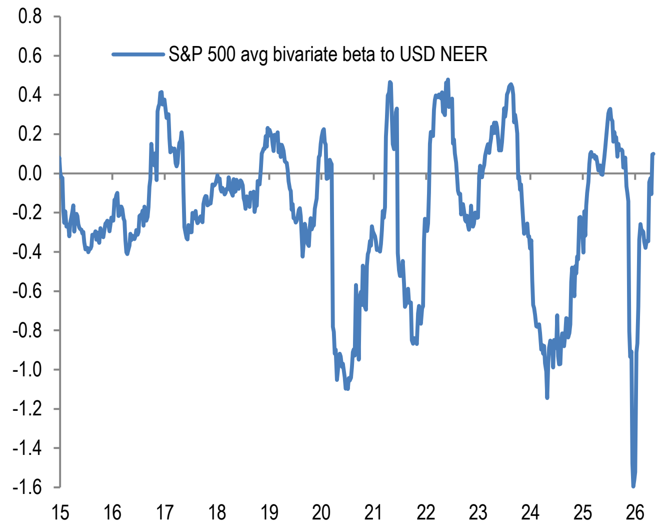
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the Eurostoxx 50 index to the weekly returns of the MSCI AC World and JPM EUR Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A48: Average beta of S&P500 companies to trade-weighted US dollar

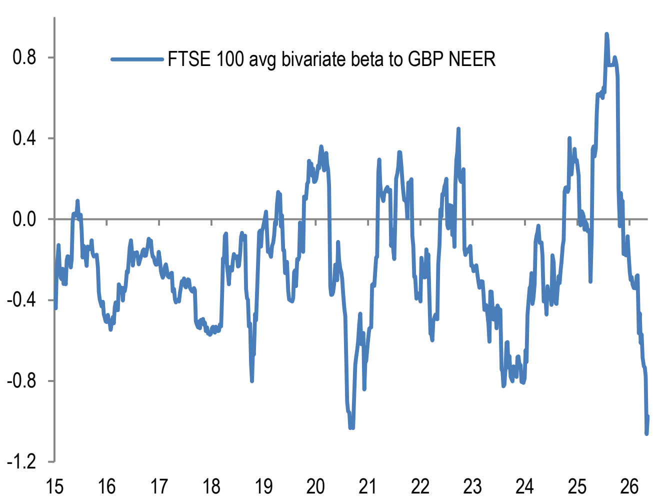
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of stocks in the S&P500 index to the weekly returns of the MSCI AC World and JPM USD Nominal broad effective exchange rate(NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A49: Average beta of FTSE 100 companies to trade-weighted GBP

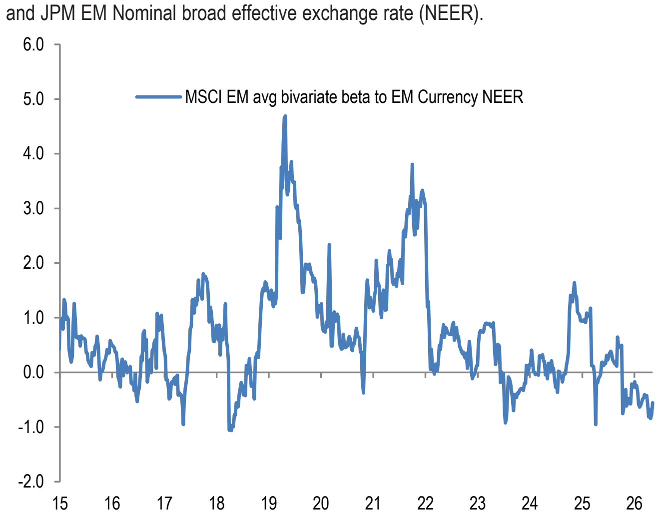
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the FTSE 100 index to the weekly returns of the MSCI AC World and JPM GBP Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A50: Average beta of MSCI EM companies to trade-weighted EM Currency Index

Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the MSCI EM index to the weekly returns of the MSCI AC World and JPM EM Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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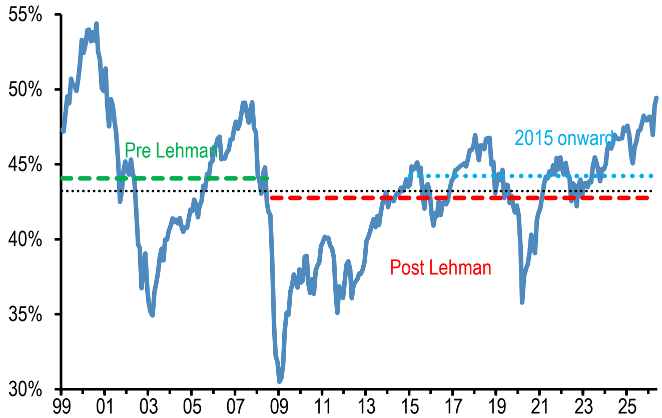
Non-Bank investors' implied allocations

Chart A51: Implied equity allocation by non-bank investors globally

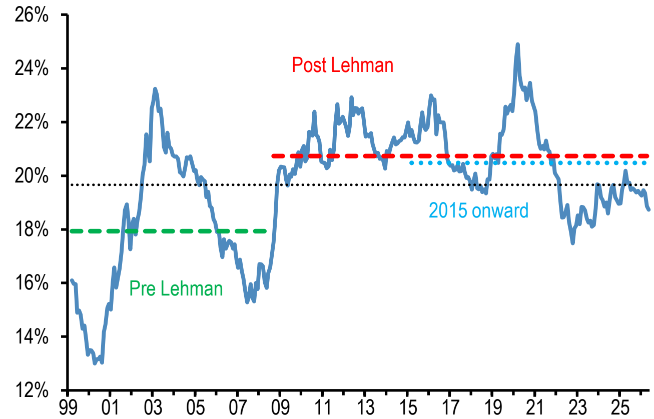
Global equities as % total holdings of equities/bonds/M2 by non-bank investors/ Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A52: Implied bond allocation by non-bank investors globally

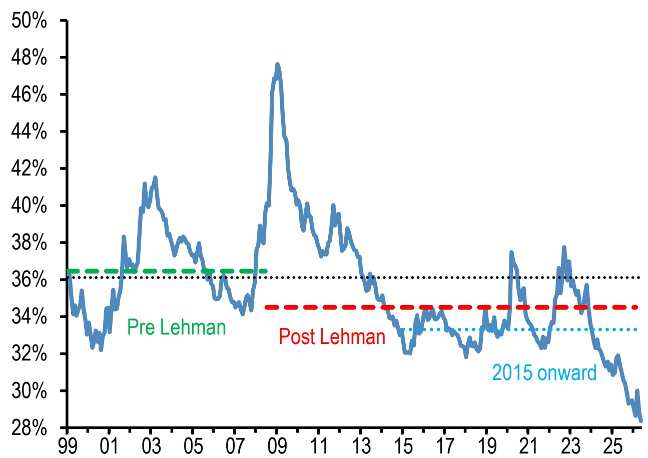
Global bonds as % total holdings of equities/bonds/M2 by non-bank investors/ Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A53: Implied cash allocation by non-bank investors globally

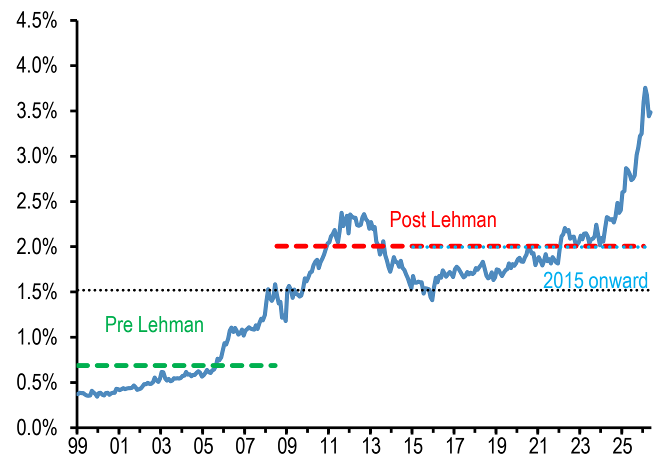
Global cash held by non-bank investors as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A54: Implied commodity allocation (including privately held gold) by non-bank investors globally

Proxied by the open interest of commodity futures ex gold and privately held gold as % of the stock of equities, bonds, cash held by non-bank investors globally and commodities including gold held by private investors.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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IB clients**	83%	79%	74%
JPMS Equity Research Coverage*	49%	39%	13%
IB clients**	94%	93%	85%

***Please note that the percentages may not add to 100% because of rounding.**
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